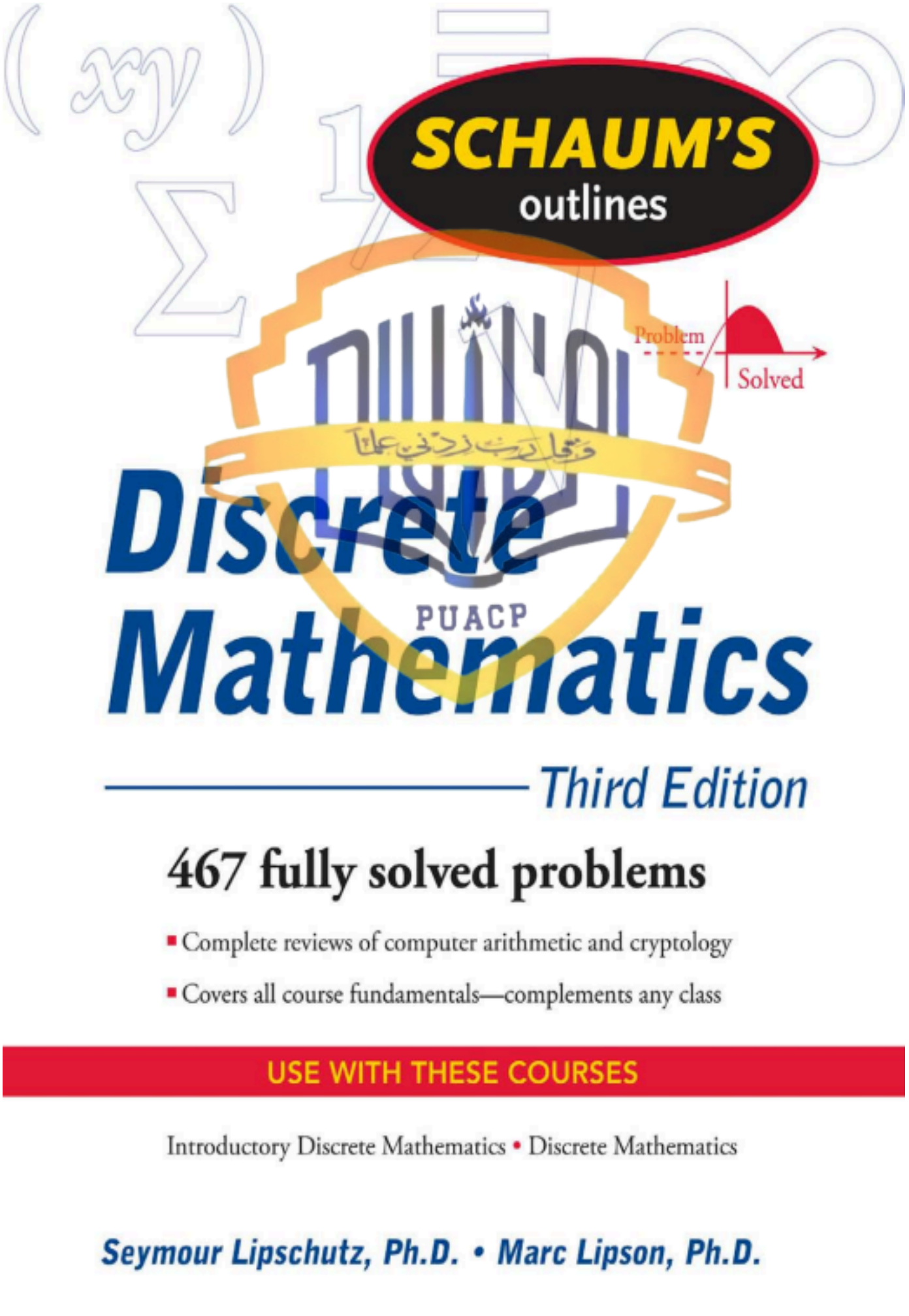




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Seymour Lipschutz, Ph.D.
Temple University

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Discrete mathematics, the study of finite systems, has become increasingly important as the computer age has advanced. The digital computer is basically a finite structure, and many of its properties can be understood and interpreted within the framework of finite mathematical systems. This book, in presenting the more essential

material, may be used as a textbook for a formal course in discrete mathematics or as a supplement to all current texts.

The first three chapters cover the standard material on sets, relations, and functions and algorithms. Next come chapters on logic, counting, and probability. We then have three chapters on graph theory: graphs, directed graphs, and binary trees. Finally there are individual chapters on properties of the integers, languages, machines, ordered sets and lattices, and Boolean algebra, and appendices on vectors and matrices, and algebraic systems. The chapter on functions and algorithms includes a discussion of cardinality and countable sets, and complexity. The chapters on graph theory include discussions on planarity, traversability, minimal paths, and Warshall's and Huffman's algorithms. We emphasize that the chapters have been written so that the order can be changed without difficulty and without loss of continuity.

Each chapter begins with a clear statement of pertinent definitions, principles, and theorems with illustrative and other descriptive material. This is followed by sets of solved and supplementary problems. The solved problems serve to illustrate and amplify the material, and also include proofs of theorems. The supplementary problems furnish a complete review of the material in the chapter. More material has been included than can be covered in most first courses. This has been done to make the book more flexible, to provide a more useful book of reference, and to stimulate further interest in the topics.

Seymour Lipschutz
Marc Lars Lipson

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Theory and Problems of DISCRE

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Set Theory

1.1 INTRODUCTION

The concept of a *set* appears in all mathematics. This chapter introduces the notation and terminology of set theory which is basic and used throughout the text. The chapter closes with the formal definition of mathematical induction, with examples.

1.2 SETS AND ELEMENTS, SUBSETS

A *set* may be viewed as any well-defined collection of objects, called the *elements* or *members* of the set. One usually uses capital letters, $A, B, X, Y, \dots$, to denote sets, and lowercase letters, $a, b, x, y, \dots$, to denote elements of sets. Synonyms for “set” are “class,” “collection,” and “family.”

Membership in a set is denoted as follows:

$a \in S$ denotes that a belongs to a set S

$a, b \in S$ denotes that a and b belong to a set S

Here $\in$ is the symbol meaning “is an element of.” We use $\notin$ to mean “is not an element of.”

Specifying Sets

There are essentially two ways to specify a particular set. One way, if possible, is to list its members separated by commas and contained in braces $\{ \}$. A second way is to state those properties which characterized the elements in the set. Examples illustrating these two ways are:

$$A = \{1, 3, 5, 7, 9\} \text{ and } B = \{x \mid x \text{ is an even integer, } x > 0\}$$

That is, A consists of the numbers 1, 3, 5, 7, 9. The second set, which reads:

B is the set of x such that x is an even integer and x is greater than 0,

denotes the set B whose elements are the positive integers. Note that a letter, usually x , is used to denote a typical member of the set; and the vertical line $|$ is read as “such that” and the comma as “and.”

EXAMPLE 1.1

(a) The set A above can also be written as $A = \{x \mid x \text{ is an odd positive integer, } x < 10\}$. (b) We cannot list all the elements of the above set B although frequently we specify the set by $B = \{2,$

4, 6, ...}

where we assume that everyone knows what we mean. Observe that $8 \in B$, but $3 \notin$

B . 1

2 SET THEORY [CHAP. 1

(c) Let $E = \{x \mid x^2 - 3x + 2 = 0\}$, $F = \{2, 1\}$ and $G = \{1, 2, 2, 1\}$. Then $E = F = G$.

We emphasize that a set does not depend on the way in which its elements are displayed. A set remains the same if its elements are repeated or rearranged.

Even if we can list the elements of a set, it may not be practical to do so. That is, we describe a set by listing its elements only if the set contains a few elements; otherwise we describe a set by the property which characterizes its elements.

Subsets

Suppose every element in a set A is also an element of a set B , that is, suppose $a \in A$ implies $a \in B$. Then A is called a *subset* of B . We also say that A is *contained* in B or that B *contains* A . This relationship is written

$$A \subseteq B \text{ or } B \supseteq A$$

Two sets are equal if they both have the same elements or, equivalently, if each is contained in the other. That is:

$$A = B \text{ if and only if } A \subseteq B \text{ and } B \subseteq A$$

If A is not a subset of B , that is, if at least one element of A does not belong to B , we write $A \not\subseteq B$.

EXAMPLE 1.2 Consider the sets:

$$A = \{1, 3, 4, 7, 8, 9\}, B = \{1, 2, 3, 4, 5\}, C = \{1, 3\}.$$

Then $C \subseteq A$ and $C \subseteq B$ since 1 and 3, the elements of C , are also members of A and B . But $B \not\subseteq A$ since some of the elements of B , e.g., 2 and 5, do not belong to A . Similarly, $A \not\subseteq B$.

Property 1: It is common practice in mathematics to put a vertical line “|” or slanted line “/” through a symbol to indicate the opposite or negative meaning of a symbol.

Property 2: The statement $A \subseteq B$ does not exclude the possibility that $A = B$. In fact, for every set A we have $A \subseteq A$ since, trivially, every element in A belongs to A . However, if $A \subseteq B$ and $A = B$, then we say A is a proper subset of B (sometimes written $A \subset B$).

Property 3: Suppose every element of a set A belongs to a set B and every element of B belongs to a set C . Then clearly every element of A also belongs to C . In other words, if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

The above remarks yield the following theorem.

Theorem 1.1: Let A, B, C be any sets. Then:

- (i) $A \subseteq A$
- (ii) If $A \subseteq B$ and $B \subseteq A$, then $A = B$
- (iii) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$

Special symbols

Some sets will occur very often in the text, and so we use special symbols for them. Some such symbols are: $\mathbf{N}$ = the set of *natural numbers* or positive integers: 1, 2, 3, ...

$\mathbf{Z}$ = the set of all integers: ..., -2, -1, 0, 1, 2, ...

$\mathbf{Q}$ = the set of rational numbers

$\mathbf{R}$ = the set of real numbers

$\mathbf{C}$ = the set of complex numbers

Observe that $\mathbf{N} \subseteq \mathbf{Z} \subseteq \mathbf{Q} \subseteq \mathbf{R} \subseteq \mathbf{C}$.

Universal Set, Empty Set

All sets under investigation in any application of set theory are assumed to belong to some fixed large set called the *universal set* which we denote by

$$U$$

unless otherwise stated or implied.

Given a universal set U and a property P , there may not be any elements of U which have property P . For example, the following set has no elements:

$$S = \{x \mid x \text{ is a positive integer, } x^2 = 3\}$$

Such a set with no elements is called the *empty set* or *null set* and is denoted by

$$\emptyset$$

There is only one empty set. That is, if S and T are both empty, then $S = T$, since they have exactly the same elements, namely, none.

The empty set $\emptyset$ is also regarded as a subset of every other set. Thus we have the following simple result which we state formally.

Theorem 1.2: For any set A , we have $\emptyset \subseteq A \subseteq U$.

Disjoint Sets

Two sets A and B are said to be *disjoint* if they have no elements in common. For example, suppose

$$A = \{1, 2\}, B = \{4, 5, 6\}, \text{ and } C = \{5, 6, 7, 8\}$$

Then A and B are disjoint, and A and C are disjoint. But B and C are not disjoint since B and C have elements in common, e.g., 5 and 6. We note that if A and B are disjoint, then neither is a subset of the other (unless one is the empty set).

1.3 VENN DIAGRAMS

A Venn diagram is a pictorial representation of sets in which sets are represented by enclosed areas in the plane. The universal set U is represented by the interior of a rectangle, and the other sets are represented by disks lying within the rectangle. If $A \subseteq B$, then the disk representing A will be entirely within the disk representing B as in Fig. 1-1(a). If A and B are disjoint, then the disk representing A will be separated from the disk representing B as in Fig. 1-1(b).

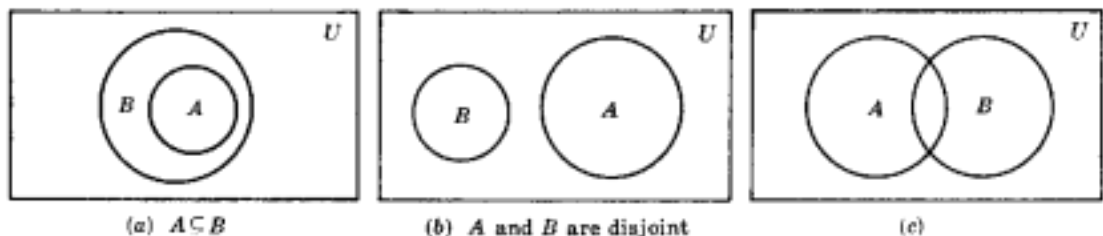


Fig. 1-1

4 SET THEORY [CHAP. 1

However, if A and B are two arbitrary sets, it is possible that some objects are in A but not in B , some are in B but not in A , some are in both A and B , and some are in neither A nor B ; hence in general we represent A and B as in Fig. 1-1(c).

Arguments and Venn Diagrams

Many verbal statements are essentially statements about sets and can therefore be described by Venn diagrams. Hence Venn diagrams can sometimes be used to determine whether or not an argument is valid.

EXAMPLE 1.3 Show that the following argument (adapted from a book on logic by Lewis Carroll, the author of *Alice in Wonderland*) is valid:

- S_1 : All my tin objects are saucepans.
 S_2 : I find all your presents very useful.
 S_3 : None of my saucepans is of the slightest use.
 S : Your presents to me are not made of tin.

The statements S_1 , S_2 , and S_3 above the horizontal line denote the assumptions, and the statement S below the line denotes the conclusion. The argument is valid if the conclusion S follows logically from the assumptions S_1 , S_2 , and S_3 .

By S_1 the tin objects are contained in the set of saucepans, and by S_3 the set of saucepans and the set of useful things are disjoint. Furthermore, by S_2 the set of “your presents” is a subset of the set of useful things. Accordingly, we can draw the Venn diagram in Fig. 1-2.

The conclusion is clearly valid by the Venn diagram because the set of “your presents” is disjoint from the set of tin objects.

Fig. 1-2

1.4 SET OPERATIONS

This section introduces a number of set operations, including the basic operations of union, intersection, and complement.

Union and Intersection

The *union* of two sets A and B , denoted by $A \cup B$, is the set of all elements which belong to A or to B ; that is,

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Here “or” is used in the sense of and/or. Figure 1-3(a) is a Venn diagram in which $A \cup B$ is shaded. The *intersection* of two sets A and B , denoted by $A \cap B$, is the set of elements which belong to both A and B ; that is,

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Figure 1-3(b) is a Venn diagram in which $A \cap B$ is shaded.

Fig. 1-3

Recall that sets A and B are said to be *disjoint* or *nonintersecting* if they have no elements in common or, using the definition of intersection, if $A \cap B = \emptyset$, the empty set. Suppose

$$S = A \cup B \text{ and } A \cap B = \emptyset$$

Then S is called the *disjoint union* of A and B .

EXAMPLE 1.4

(a) Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{2, 3, 8, 9\}$. Then

$$\begin{aligned} A \cup B &= \{1, 2, 3, 4, 5, 6, 7\}, A \cup C = \{1, 2, 3, 4, 8, 9\}, B \cup C = \{2, 3, 4, 5, 6, 7, 8, 9\}, \\ A \cap B &= \{3, 4\}, A \cap C = \{2, 3\}, B \cap C = \{3\}. \end{aligned}$$

(b) Let U be the set of students at a university, and let M denote the set of male students and let F denote the set of female students. The U is the disjoint union of M and F ; that is,

$$U = M \cup F \text{ and } M \cap F = \emptyset$$

This comes from the fact that every student in U is either in M or in F , and clearly no student belongs to both M and F , that is, M and F are disjoint.

The following properties of union and intersection should be noted.

Property 1: Every element x in $A \cap B$ belongs to both A and B ; hence x belongs to A and x belongs to B . Thus $A \cap B$ is a subset of A and of B ; namely

$$A \cap B \subseteq A \text{ and } A \cap B \subseteq B$$

Property 2: An element x belongs to the union $A \cup B$ if x belongs to A or x belongs to B ; hence every element in A belongs to $A \cup B$, and every element in B belongs to $A \cup B$. That is,

$$A \subseteq A \cup B \text{ and } B \subseteq A \cup B$$

We state the above results formally:

Theorem 1.3: For any sets A and B , we have:

$$(i) A \cap B \subseteq A \subseteq A \cup B \text{ and } (ii) A \cap B \subseteq B \subseteq A \cup B.$$

The operation of set inclusion is closely related to the operations of union and intersection, as shown by the following theorem.

Theorem 1.4: The following are equivalent: $A \subseteq B$, $A \cap B = A$, $A \cup B = B$.

This theorem is proved in Problem 1.8. Other equivalent conditions to are given in Problem 1.31.

Fig. 1-4

Complements, Differences, Symmetric Differences

Recall that all sets under consideration at a particular time are subsets of a fixed universal set U . The *absolute complement* or, simply, *complement* of a set A , denoted by A^C , is the set of elements which belong to U but which do not belong to A . That is,

$$A^C = \{x \mid x \in U, x \notin A\}$$

Some texts denote the complement of A by $\bar{A}$ or A' . Fig. 1-4(a) is a Venn diagram in which A^C is shaded.

The *relative complement* of a set B with respect to a set A or, simply, the *difference* of A and B , denoted by $A \setminus B$, is the set of elements which belong to A but which do not belong to B ; that is

$$A \setminus B = \{x \mid x \in A, x \notin B\}$$

The set $A \setminus B$ is read “ A minus B .” Many texts denote $A \setminus B$ by $A - B$ or $A \sim B$. Fig. 1-4(b) is a Venn diagram in which $A \setminus B$ is shaded.

The *symmetric difference* of sets A and B , denoted by $A \oplus B$, consists of those elements which belong to A or B but not to both. That is,

$$A \oplus B = (A \cup B) \setminus (A \cap B) \text{ or } A \oplus B = (A \setminus B) \cup (B \setminus A)$$

Figure 1-4(c) is a Venn diagram in which $A \oplus B$ is shaded.

EXAMPLE 1.5 Suppose $U = N = \{1, 2, 3, \dots\}$ is the universal set. Let

$$A = \{1, 2, 3, 4\}, B = \{3, 4, 5, 6, 7\}, C = \{2, 3, 8, 9\}, E = \{2, 4, 6, \dots\}$$

(Here E is the set of even integers.) Then:

$$A^C = \{5, 6, 7, \dots\}, B^C = \{1, 2, 8, 9, 10, \dots\}, E^C = \{1, 3, 5, 7, \dots\}$$

That is, E^C is the set of odd positive integers. Also:

$$\begin{aligned} A \setminus B &= \{1, 2\}, A \setminus C = \{1, 4\}, B \setminus C = \{4, 5, 6, 7\}, A \setminus E = \{1, 3\}, \\ B \setminus A &= \{5, 6, 7\}, C \setminus A = \{8, 9\}, C \setminus B = \{2, 8, 9\}, E \setminus A = \{6, 8, 10, 12, \dots\}. \end{aligned}$$

Furthermore:

$$\begin{aligned} A \oplus B &= (A \setminus B) \cup (B \setminus A) = \{1, 2, 5, 6, 7\}, B \oplus C = \{2, 4, 5, 6, 7, 8, 9\}, \\ A \oplus C &= (A \setminus C) \cup (C \setminus A) = \{1, 4, 8, 9\}, A \oplus E = \{1, 3, 6, 8, 10, \dots\}. \end{aligned}$$

Fundamental Products

Consider n distinct sets $A_1, A_2, \dots, A_n$. A *fundamental product* of the sets is a set of the form

$$A_1^* \cap A_2^* \cap \dots \cap A_n^* \text{ where } A_i^* = A_i \text{ or } A_i^* = A_i^C$$

We note that:

- (i) There are $m = 2^n$ such fundamental products.

- (ii) Any two such fundamental products are disjoint.
 (iii) The universal set **U** is the union of all fundamental products.

Thus **U** is the disjoint union of the fundamental products (Problem 1.60). There is a geometrical description of these sets which is illustrated below.

EXAMPLE 1.6 Figure 1-5(a) is the Venn diagram of three sets **A**, **B**, **C**. The following lists the $m = 2^3 = 8$ fundamental products of the sets **A**, **B**, **C**:

$$P_1 = A \cap B \cap C, P_3 = A \cap B^C \cap C, P_5 = A^C \cap B \cap C, P_7 = A^C \cap B^C \cap C, P_2 = A \cap B \cap C^C, P_4 = A \cap B^C \cap C^C, P_6 = A^C \cap B \cap C^C, P_8 = A^C \cap B^C \cap C^C.$$

The eight products correspond precisely to the eight disjoint regions in the Venn diagram of sets **A**, **B**, **C** as indicated by the labeling of the regions in Fig. 1-5(b).

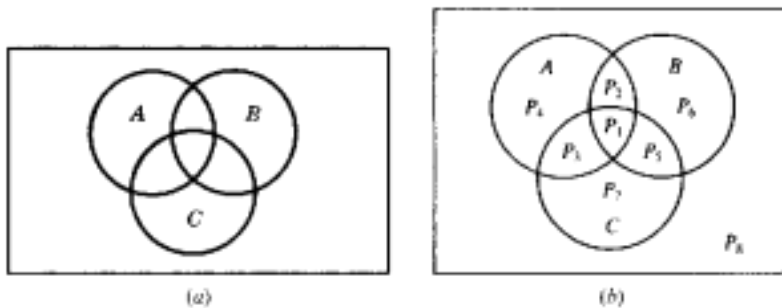


Fig. 1-5

1.5 ALGEBRA OF SETS, DUALITY

Sets under the operations of union, intersection, and complement satisfy various laws (identities) which are listed in Table 1-1. In fact, we formally state this as:

Theorem 1.5: Sets satisfy the laws in Table 1-1.

Table 1-1 Laws of the algebra of sets

		Associative laws: (2a) $(A \cup B) \cup C = A \cup (B \cup C)$
		(2b) $(A \cap B) \cap C = A \cap (B \cap C)$
		Commutative laws: (3a) $A \cup B = B \cup A$ (3b) $A \cap B = B \cap A$
		Distributive laws: (4a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (4b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
		Identity laws: (5a) $A \cup \emptyset = A$ (5b) $A \cap U = A$ (6a) $A \cup U = U$ (6b) $A \cap \emptyset = \emptyset$
		Involution laws: (7) $(A^C)^C = A$
		Complement laws: (8a) $A \cup A^C = U$ (8b) $A \cap A^C = \emptyset$ (9a) $U^C = \emptyset$ (9b) $\emptyset^C = U$
		DeMorgan's laws: (10a) $(A \cup B)^C = A^C \cap B^C$ (10b) $(A \cap B)^C = A^C \cup B^C$

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Remark: Each law in Table 1-1 follows from an equivalent logical law. Consider, for example, the proof of DeMorgan's Law 10(a):

$$(A \cup B)^C = \{x \mid x \notin (A \text{ or } B)\} = \{x \mid x \notin A \text{ and } x \notin B\} = A^C \cap B^C$$

Here we use the equivalent (DeMorgan's) logical law:

$$\neg(p \vee q) = \neg p \wedge \neg q$$

where $\neg$ means "not," $\vee$ means "or," and $\wedge$ means "and." (Sometimes Venn diagrams are used to illustrate the laws in Table 1-1 as in Problem 1.17.)

Duality

The identities in Table 1-1 are arranged in pairs, as, for example, (2a) and (2b). We now consider the principle behind this arrangement. Suppose E is an equation of set algebra. The dual E^* of E is the equation obtained by replacing each occurrence of $\cup$, $\cap$, $\bar{}$ and $\emptyset$ in E by $\cap$, $\cup$, $\emptyset$, and $\bar{}$, respectively. For example, the dual of

$$(\bar{A} \cap B) \cup (A \cap B) = B \text{ is } (\emptyset \cup A) \cap (B \cup A) = A$$

Observe that the pairs of laws in Table 1-1 are duals of each other. It is a fact of set algebra, called the *principle of duality*, that if any equation E is an identity then its dual E^* is also an identity.

1.6 FINITE SETS, COUNTING PRINCIPLE

Sets can be finite or infinite. A set S is said to be *finite* if S is empty or if S contains exactly m elements where m is a positive integer; otherwise S is *infinite*.

EXAMPLE 1.7

- (a) The set A of the letters of the English alphabet and the set D of the days of the week are finite sets. Specifically, A has 26 elements and D has 7 elements.
- (b) Let E be the set of even positive integers, and let I be the *unit interval*, that is,

$$E = \{2, 4, 6, \dots\} \text{ and } I = [0, 1] = \{x \mid 0 \leq x \leq 1\}$$

Then both E and I are infinite.

A set S is *countable* if S is finite or if the elements of S can be arranged as a sequence, in which case S is said to be *countably infinite*; otherwise S is said to be *uncountable*. The above set E of even integers is countably infinite, whereas one can prove that the unit interval $I = [0, 1]$ is uncountable.

Counting Elements in Finite Sets

The notation $n(S)$ or $|S|$ will denote the number of elements in a set S . (Some texts use $\#(S)$ or $\text{card}(S)$ instead of $n(S)$.) Thus $n(A) = 26$, where A is the letters in the English alphabet, and $n(D) = 7$, where D is the days of the week. Also $n(\emptyset) = 0$ since the empty set has no elements.

The following lemma applies.

Lemma 1.6: Suppose A and B are finite disjoint sets. Then $A \cup B$ is finite and

$$n(A \cup B) = n(A) + n(B)$$

This lemma may be restated as follows:

Lemma 1.6: Suppose S is the disjoint union of finite sets A and B . Then S is finite and

$$n(S) = n(A) + n(B)$$

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Proof. In counting the elements of $A \cup B$, first count those that are in A . There are $n(A)$ of these. The only other elements of $A \cup B$ are those that are in B but not in A . But since A and B are disjoint, no element of B is in A , so there are $n(B)$ elements that are in B but not in A . Therefore, $n(A \cup B) = n(A) + n(B)$.

For any sets A and B , the set A is the disjoint union of $A \setminus B$ and $A \cap B$. Thus Lemma 1.6 gives us the following useful result.

Corollary 1.7: Let A and B be finite sets. Then

$$n(A \setminus B) = n(A) - n(A \cap B)$$

For example, suppose an art class A has 25 students and 10 of them are taking a biology class B . Then the number of students in class A which are not in class B is:

$$n(A \setminus B) = n(A) - n(A \cap B) = 25 - 10 = 15$$

Given any set A , recall that the universal set U is the disjoint union of A and A^C . Accordingly, Lemma 1.6 also gives the following result.

Corollary 1.8: Let A be a subset of a finite universal set U . Then

$$n(A^C) = n(U) - n(A)$$

For example, suppose a class U with 30 students has 18 full-time students. Then there are $30 - 18 = 12$ part-time students in the class U .

Inclusion–Exclusion Principle

There is a formula for $n(A \cup B)$ even when they are not disjoint, called the Inclusion–Exclusion Principle. Namely:

Theorem (Inclusion–Exclusion Principle) 1.9: Suppose A and B are finite sets. Then $A \cup B$ and $A \cap B$ are finite and

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

That is, we find the number of elements in A or B (or both) by first adding $n(A)$ and $n(B)$ (inclusion) and then subtracting $n(A \cap B)$ (exclusion) since its elements were counted twice.

We can apply this result to obtain a similar formula for three sets:

Corollary 1.10: Suppose A, B, C are finite sets. Then $A \cup B \cup C$ is finite and

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Mathematical induction (Section 1.8) may be used to further generalize this result to any number of finite sets.

EXAMPLE 1.8 Suppose a list A contains the 30 students in a mathematics class, and a list B contains the 35 students in an English class, and suppose there are 20 names on both lists. Find the number of students: (a) only on list A , (b) only on list B , (c) on list A or B (or both), (d) on exactly one list.

(a) List A has 30 names and 20 are on list B ; hence $30 - 20 = 10$ names are only on list A .

(b) Similarly, $35 - 20 = 15$ are only on list B .

(c) We seek $n(A \cup B)$. By inclusion–exclusion,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 30 + 35 - 20 = 45.$$

In other words, we combine the two lists and then cross out the 20 names which appear twice.

(d) By (a) and (b), $10 + 15 = 25$ names are only on one list; that is, $n(A \oplus B) = 25$.

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1.7 CLASSES OF SETS, POWER SETS, PARTITIONS

Given a set S , we might wish to talk about some of its subsets. Thus we would be considering a *set of sets*. Whenever such a situation occurs, to avoid confusion, we will speak of a *class* of sets or *collection* of sets rather than a *set* of sets. If we wish to consider some of the sets in a given class of sets, then we speak of *subclass* or *subcollection*.

EXAMPLE 1.9 Suppose $S = \{1, 2, 3, 4\}$.

(a) Let A be the class of subsets of S which contain exactly three elements of S . Then

$$A = [\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}]$$

That is, the elements of A are the sets $\{1, 2, 3\}$, $\{1, 2, 4\}$, $\{1, 3, 4\}$, and $\{2, 3, 4\}$.

(b) Let B be the class of subsets of S , each which contains 2 and two other elements of S . Then

$$B = [\{1, 2, 3\}, \{1, 2, 4\}, \{2, 3, 4\}]$$

The elements of B are the sets $\{1, 2, 3\}$, $\{1, 2, 4\}$, and $\{2, 3, 4\}$. Thus B is a subclass of A , since every element of B is also an element of A . (To avoid confusion, we will sometimes enclose the sets of a class in

brackets instead of braces.)

Power Sets

For a given set S , we may speak of the class of all subsets of S . This class is called the *power set* of S , and will be denoted by $P(S)$. If S is finite, then so is $P(S)$. In fact, the number of elements in $P(S)$ is 2 raised to the power $n(S)$. That is,

$$n(P(S)) = 2^{n(S)}$$

(For this reason, the power set of S is sometimes denoted by 2^S .)

EXAMPLE 1.10 Suppose $S = \{1, 2, 3\}$. Then

$$P(S) = [\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, S]$$

Note that the empty set $\emptyset$ belongs to $P(S)$ since $\emptyset$ is a subset of S . Similarly, S belongs to $P(S)$. As expected from the above remark, $P(S)$ has $2^3 = 8$ elements.

Partitions

Let S be a nonempty set. A *partition* of S is a subdivision of S into nonoverlapping, nonempty subsets. Precisely, a *partition* of S is a collection $\{A_i\}$ of nonempty subsets of S such that:

- (i) Each a in S belongs to one of the A_i .
- (ii) The sets of $\{A_i\}$ are mutually disjoint; that is, if

$$A_j = A_k \text{ then } A_j \cap A_k = \emptyset$$

The subsets in a partition are called *cells*. Figure 1-6 is a Venn diagram of a partition of the rectangular set S of points into five cells, A_1, A_2, A_3, A_4, A_5 .



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Fig. 1-6

EXAMPLE 1.11 Consider the following collections of subsets of $S = \{1, 2, \dots, 8, 9\}$:

- (i) $\{\{1, 3, 5\}, \{2, 6\}, \{4, 8, 9\}\}$
- (ii) $\{\{1, 3, 5\}, \{2, 4, 6, 8\}, \{5, 7, 9\}\}$
- (iii) $\{\{1, 3, 5\}, \{2, 4, 6, 8\}, \{7, 9\}\}$

Then (i) is not a partition of S since 7 in S does not belong to any of the subsets. Furthermore, (ii) is not a partition of S since $\{1, 3, 5\}$ and $\{5, 7, 9\}$ are not disjoint. On the other hand, (iii) is a partition of S .

Generalized Set Operations

The set operations of union and intersection were defined above for two sets. These operations can be extended to any number of sets, finite or infinite, as follows.

Consider first a finite number of sets, say, $A_1, A_2, \dots, A_m$. The union and intersection of these sets are denoted and defined, respectively, by

$$A_1 \cup A_2 \cup \dots \cup A_m = \bigcup_{i=1}^m A_i = \{x \mid x \in A_i \text{ for some } A_i\}$$

$$A_1 \cap A_2 \cap \dots \cap A_m = \bigcap_{i=1}^m A_i = \{x \mid x \in A_i \text{ for every } A_i\}$$

That is, the union consists of those elements which belong to at least one of the sets, and the intersection consists of those elements which belong to all the sets.

Now let $\mathcal{A}$ be any collection of sets. The union and the intersection of the sets in the collection $\mathcal{A}$ is denoted and defined, respectively, by

$$(A \mid A \in \mathcal{A}) = \{x \mid x \in A_i \text{ for some } A_i \in \mathcal{A}\}$$

$$(A \mid A \in \mathcal{A}) = \{x \mid x \in A_i \text{ for every } A_i \in \mathcal{A}\}$$

That is, the union consists of those elements which belong to at least one of the sets in the collection $\mathcal{A}$ and the intersection consists of those elements which belong to every set in the collection $\mathcal{A}$.

EXAMPLE 1.12 Consider the sets

$$A_1 = \{1, 2, 3, \dots\} = \mathbf{N}, A_2 = \{2, 3, 4, \dots\}, A_3 = \{3, 4, 5, \dots\}, A_n = \{n, n+1, n+2, \dots\}.$$

Then the union and intersection of the sets are as follows:

$$(A_k \mid k \in \mathbf{N}) = \mathbf{N} \text{ and } (A_k \mid k \in \mathbf{N}) = \emptyset$$

DeMorgan's laws also hold for the above generalized operations. That is:

Theorem 1.11: Let $\mathcal{A}$ be a collection of sets. Then:

$$(i) \quad (A \mid A \in \mathcal{A})^C = (A^C \mid A \in \mathcal{A})$$

$$(ii) \quad (A \mid A \in \mathcal{A})^C = (A^C \mid A \in \mathcal{A})$$

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1.8 MATHEMATICAL INDUCTION

An essential property of the set $\mathbf{N} = \{1, 2, 3, \dots\}$ of positive integers follows:

Principle of Mathematical Induction I: Let P be a proposition defined on the positive integers $\mathbf{N}$; that is, $P(n)$ is either true or false for each $n \in \mathbf{N}$. Suppose P has the following two properties:

(i) $P(1)$ is true.

(ii) $P(k+1)$ is true whenever $P(k)$ is true.

Then P is true for every positive integer $n \in \mathbf{N}$.

We shall not prove this principle. In fact, this principle is usually given as one of the axioms when $\mathbf{N}$ is developed axiomatically.

EXAMPLE 1.13 Let P be the proposition that the sum of the first n odd numbers is n^2 ; that is,

$$P(n) : 1 + 3 + 5 + \dots + (2n-1) = n^2$$

(The k th odd number is $2k-1$, and the next odd number is $2k+1$.) Observe that $P(n)$ is true for $n=1$; namely,

$$P(1) = 1^2$$

Assuming $P(k)$ is true, we add $2k+1$ to both sides of $P(k)$, obtaining

$$1 + 3 + 5 + \dots + (2k-1) + (2k+1) = k^2 + (2k+1) = (k+1)^2$$

which is $P(k+1)$. In other words, $P(k+1)$ is true whenever $P(k)$ is true. By the principle of mathematical induction, P is true for all n .

There is a form of the principle of mathematical induction which is sometimes more convenient to use. Although it appears different, it is really equivalent to the above principle of induction.

Principle of Mathematical Induction II: Let P be a proposition defined on the positive integers $\mathbf{N}$ such that:

(i) $P(1)$ is true.

(ii) $P(k)$ is true whenever $P(j)$ is true for all $1 \leq j < k$.
Then P is true for every positive integer $n \in \mathbf{N}$.

Remark: Sometimes one wants to prove that a proposition P is true for the set of integers

$$\{a, a + 1, a + 2, a + 3, \dots\}$$

where a is any integer, possibly zero. This can be done by simply replacing 1 by a in either of the above Principles of Mathematical Induction.

Solved Problems

SETS AND SUBSETS

1.1 Which of these sets are equal: $\{x, y, z\}$, $\{z, y, z, x\}$, $\{y, x, y, z\}$, $\{y, z, x, y\}$?

They are all equal. Order and repetition do not change a set.

1.2 List the elements of each set where $\mathbf{N} = \{1, 2, 3, \dots\}$.

(a) $A = \{x \in \mathbf{N} \mid 3 < x < 9\}$

(b) $B = \{x \in \mathbf{N} \mid x \text{ is even, } x < 11\}$

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(c) $C = \{x \in \mathbf{N} \mid 4 + x = 3\}$

(a) A consists of the positive integers between 3 and 9; hence $A = \{4, 5, 6, 7, 8\}$.

(b) B consists of the even positive integers less than 11; hence $B = \{2, 4, 6, 8, 10\}$.

(c) No positive integer satisfies $4 + x = 3$; hence $C = \emptyset$, the empty set.

1.3 Let $A = \{2, 3, 4, 5\}$.

(a) Show that A is not a subset of $B = \{x \in \mathbf{N} \mid x \text{ is even}\}$.

(b) Show that A is a proper subset of $C = \{1, 2, 3, \dots, 8, 9\}$.

(a) It is necessary to show that at least one element in A does not belong to B . Now $3 \in A$ and, since B consists of even numbers, $3 \notin B$; hence A is not a subset of B .

(b) Each element of A belongs to C so $A \subseteq C$. On the other hand, $1 \in C$ but $1 \notin A$. Hence $A \subset C$. Therefore A is a proper subset of C .

SET OPERATIONS

1.4 Let $\mathbf{U} = \{1, 2, \dots, 9\}$ be the universal set, and let

$$A = \{1, 2, 3, 4, 5\}, C = \{5, 6, 7, 8, 9\}, E = \{2, 4, 6, 8\}, \\ B = \{4, 5, 6, 7\}, D = \{1, 3, 5, 7, 9\}, F = \{1, 5, 9\}.$$

Find: (a) $A \cup B$ and $A \cap B$; (b) $A \cup C$ and $A \cap C$; (c) $D \cup F$ and $D \cap F$.

Recall that the union $X \cup Y$ consists of those elements in either X or Y (or both), and that the intersection $X \cap Y$ consists of those elements in both X and Y .

(a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$ and $A \cap B = \{4, 5\}$

(b) $A \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} = \mathbf{U}$ and $A \cap C = \{5\}$

(c) $D \cup F = \{1, 3, 5, 7, 9\} = D$ and $D \cap F = \{1, 5, 9\} = F$

Observe that $F \subseteq D$, so by Theorem 1.4 we must have $D \cup F = D$ and $D \cap F = F$.

1.5 Consider the sets in the preceding Problem 1.4. Find:

(a) A^C, B^C, D^C, E^C ; (b) $A \setminus B, B \setminus A, D \setminus E$; (c) $A \oplus B, C \oplus D, E \oplus F$.

Recall that:

(1) The complements X^C consists of those elements in $\mathbf{U}$ which do not belong to X .

(2) The difference $X \setminus Y$ consists of the elements in X which do not belong to Y .

(3) The symmetric difference $X \oplus Y$ consists of the elements in X or in Y but not in both.

Therefore:

(a) $A^C = \{6, 7, 8, 9\}$; $B^C = \{1, 2, 3, 8, 9\}$; $D^C = \{2, 4, 6, 8\} = E$; $E^C = \{1, 3, 5, 7, 9\} = D$. (b) $A \setminus B = \{1, 2, 3\}$; $B \setminus A = \{6, 7\}$; $D \setminus E = \{1, 3, 5, 7, 9\} = D$; $F \setminus D = \emptyset$.
(c) $A \oplus B = \{1, 2, 3, 6, 7\}$; $C \oplus D = \{1, 3, 6, 8\}$; $E \oplus F = \{2, 4, 6, 8, 1, 5, 9\} = E \cup F$.

1.6 Show that we can have: (a) $A \cap B = A \cap C$ without $B = C$; (b) $A \cup B = A \cup C$ without $B = C$.

(a) Let $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{2, 4\}$. Then $A \cap B = \{2\}$ and $A \cap C = \{2\}$; but $B \neq C$. (b) Let $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{2, 3\}$. Then $A \cup B = \{1, 2, 3\}$ and $A \cup C = \{1, 2, 3\}$ but $B \neq C$.

1.7 Prove: $B \setminus A = B \cap A^C$. Thus, the set operation of difference can be written in terms of the operations of intersection and complement.

$$B \setminus A = \{x \mid x \in B, x \notin A\} = \{x \mid x \in B, x \in A^C\} = B \cap A^C.$$

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1.8 Prove Theorem 1.4. The following are equivalent: $A \subseteq B$, $A \cap B = A$, $A \cup B = B$.

Suppose $A \subseteq B$ and let $x \in A$. Then $x \in B$, hence $x \in A \cap B$ and $A \subseteq A \cap B$. By Theorem 1.3, $(A \cap B) \subseteq A$. Therefore $A \cap B = A$. On the other hand, suppose $A \cap B = A$ and let $x \in A$. Then $x \in (A \cap B)$; hence $x \in A$ and $x \in B$. Therefore, $A \subseteq B$. Both results show that $A \subseteq B$ is equivalent to $A \cap B = A$.

Suppose again that $A \subseteq B$. Let $x \in (A \cup B)$. Then $x \in A$ or $x \in B$. If $x \in A$, then $x \in B$ because $A \subseteq B$. In either case, $x \in B$. Therefore $A \cup B \subseteq B$. By Theorem 1.3, $B \subseteq A \cup B$. Therefore $A \cup B = B$. Now suppose $A \cup B = B$ and let $x \in A$. Then $x \in A \cup B$ by definition of the union of sets. Hence $x \in B = A \cup B$. Therefore $A \subseteq B$. Both results show that $A \subseteq B$ is equivalent to $A \cup B = B$.

Thus $A \subseteq B$, $A \cap B = A$ and $A \cup B = B$ are equivalent.

VENN DIAGRAMS, ALGEBRA OF SETS, DUALITY

1.9 Illustrate DeMorgan's Law $(A \cup B)^C = A^C \cap B^C$ using Venn diagrams.

Shade the area outside $A \cup B$ in a Venn diagram of sets A and B . This is shown in Fig. 1-7(a); hence the shaded area represents $(A \cup B)^C$. Now shade the area outside A in a Venn diagram of A and B with strokes in one direction (////), and then shade the area outside B with strokes in another direction (\\\\). This is shown in Fig. 1-7(b); hence the cross-hatched area (area where both lines are present) represents $A^C \cap B^C$. Both $(A \cup B)^C$ and $A^C \cap B^C$ are represented by the same area; thus the Venn diagram indicates $(A \cup B)^C = A^C \cap B^C$. (We emphasize that a Venn diagram is not a formal proof, but it can indicate relationships between sets.)

Fig. 1-7

1.10 Prove the Distributive Law: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

$$\begin{aligned} A \cap (B \cup C) &= \{x \mid x \in A, x \in (B \cup C)\} \\ &= \{x \mid x \in A, x \in B \text{ or } x \in A, x \in C\} = (A \cap B) \cup (A \cap C) \end{aligned}$$

Here we use the analogous logical law $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ where $\wedge$ denotes “and” and $\vee$ denotes “or.”

1.11 Write the dual of: (a) $(U \cap A) \cup (B \cap A) = A$; (b) $(A \cap U) \cap (\emptyset \cup A^C) = \emptyset$.

Interchange $\cup$ and $\cap$ and also U and $\emptyset$ in each set equation:

(a) $(\emptyset \cup A) \cap (B \cup A) = A$; (b) $(A \cup \emptyset) \cup (U \cap A^C) = U$.

1.12 Prove: $(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$. (Thus either one may be used to define $A \oplus B$.)

Using $X \setminus Y = X \cap Y^C$ and the laws in Table 1.1, including DeMorgan's Law, we obtain:

$$\begin{aligned} (A \cup B) \setminus (A \cap B) &= (A \cup B) \cap (A \cap B)^C = (A \cup B) \cap (A^C \cup B^C) \\ &= (A \cup A^C) \cup (A \cap B^C) \cup (B \cap A^C) \cup (B \cap B^C) \\ &= \emptyset \cup (A \cap B^C) \cup (B \cap A^C) \cup \emptyset \\ &= (A \cap B^C) \cup (B \cap A^C) = (A \setminus B) \cup (B \setminus A) \end{aligned}$$

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1.13 Determine the validity of the following argument:

S_1 : All my friends are musicians.

S_2 : John is my friend.

S_3 : None of my neighbors are musicians.

S : John is not my neighbor.

The premises S_1 and S_3 lead to the Venn diagram in Fig. 1-8(a). By S_2 , John belongs to the set of friends which is disjoint from the set of neighbors. Thus S is a valid conclusion and so the argument is valid.

Fig. 1-8

FINITE SETS AND THE COUNTING PRINCIPLE

1.14 Each student in Liberal Arts at some college has a mathematics requirement A and a science requirement B . A poll of 140 sophomore students shows that:

60 completed A , 45 completed B , 20 completed both A and B .

Use a Venn diagram to find the number of students who have completed:

(a) At least one of A and B ; (b) exactly one of A or B ; (c) neither A nor B .

Translating the above data into set notation yields:

$$n(A) = 60, n(B) = 45, n(A \cap B) = 20, n(U) = 140$$

Draw a Venn diagram of sets A and B as in Fig. 1-1(c). Then, as in Fig. 1-8(b), assign numbers to the four regions as follows:

20 completed both A and B , so $n(A \cap B) = 20$.

$60 - 20 = 40$ completed A but not B , so $n(A \setminus B) = 40$.

$45 - 20 = 25$ completed B but not A , so $n(B \setminus A) = 25$.

$140 - 20 - 40 - 25 = 55$ completed neither A nor B .

By the Venn diagram:

(a) $20 + 40 + 25 = 85$ completed A or B . Alternately, by the Inclusion–Exclusion Principle:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 60 + 45 - 20 = 85$$

(b) $40 + 25 = 65$ completed exactly one requirement. That is, $n(A \oplus B) = 65$.

(c) 55 completed neither requirement, i.e. $n(A^C \cap B^C) = n[(A \cup B)^C] = 140 - 85 = 55$.

1.15 In a survey of 120 people, it was found that:

65 read *Newsweek* magazine, 20 read both *Newsweek* and *Time*,

45 read *Time*, 25 read both *Newsweek* and *Fortune*,

42 read *Fortune*, 15 read both *Time* and *Fortune*,

8 read all three magazines.

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(a) Find the number of people who read at least one of the three magazines.

(b) Fill in the correct number of people in each of the eight regions of the Venn diagram in Fig. 1-9(a) where N , T , and F denote the set of people who read *Newsweek*, *Time*, and *Fortune*, respectively.

(c) Find the number of people who read exactly one magazine.

Fig. 1-9

(a) We want to find $n(N \cup T \cup F)$. By Corollary 1.10 (Inclusion–Exclusion Principle),

$$\begin{aligned} n(N \cup T \cup F) &= n(N) + n(T) + n(F) - n(N \cap T) - n(N \cap F) - n(T \cap F) + n(N \cap T \cap F) \\ &= 65 + 45 + 42 - 20 - 25 - 15 + 8 = 100 \end{aligned}$$

(b) The required Venn diagram in Fig. 1-9(b) is obtained as follows:

8 read all three magazines,

$20 - 8 = 12$ read *Newsweek* and *Time* but not all three magazines,

$25 - 8 = 17$ read *Newsweek* and *Fortune* but not all three magazines,

$15 - 8 = 7$ read *Time* and *Fortune* but not all three magazines,

$65 - 12 - 8 - 17 = 28$ read only *Newsweek*,

$45 - 12 - 8 - 7 = 18$ read only *Time*,

$42 - 17 - 8 - 7 = 10$ read only *Fortune*,

$120 - 100 = 20$ read no magazine at all.

(c) $28 + 18 + 10 = 56$ read exactly one of the magazines.

1.16 Prove Theorem 1.9. Suppose A and B are finite sets. Then $A \cup B$ and $A \cap B$ are finite and

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

If A and B are finite then, clearly, $A \cup B$ and $A \cap B$ are finite.

Suppose we count the elements in A and then count the elements in B .

Then every element in $A \cap B$ would be counted twice, once in A and once in B . Thus

CLASSES OF SETS

1.17 Let $A = [\{1, 2, 3\}, \{4, 5\}, \{6, 7, 8\}]$. (a) List the elements of A ; (b) Find $n(A)$.

(a) A has three elements, the sets $\{1, 2, 3\}$, $\{4, 5\}$, and $\{6, 7, 8\}$.

(b) $n(A) = 3$.

1.18 Determine the power set $P(A)$ of $A = \{a, b, c, d\}$.

The elements of $P(A)$ are the subsets of A . Hence

$$P(A) = [A, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a\}, \{b\}, \{c\}, \{d\}, \emptyset]$$

As expected, $P(A)$ has $2^4 = 16$ elements.

1.19 Let $S = \{a, b, c, d, e, f, g\}$. Determine which of the following are partitions of S :

(a) $P_1 = [\{a, c, e\}, \{b\}, \{d, g\}]$, (c) $P_3 = [\{a, b, e, g\}, \{c\}, \{d, f\}]$,

(b) $P_2 = [\{a, e, g\}, \{c, d\}, \{b, f\}]$, (d) $P_4 = [\{a, b, c, d, e, f, g\}]$.

(a) P_1 is not a partition of S since $f \in S$ does not belong to any of the cells.

(b) P_2 is not a partition of S since $e \in S$ belongs to two of the cells.

(c) P_3 is a partition of S since each element in S belongs to exactly one cell.

(d) P_4 is a partition of S into one cell, S itself.

1.20 Find all partitions of $S = \{a, b, c, d\}$.

Note first that each partition of S contains either 1, 2, 3, or 4 distinct cells. The partitions are as follows:

(1) $[\{a, b, c, d\}]$

(2) $[\{a\}, \{b, c, d\}], [\{b\}, \{a, c, d\}], [\{c\}, \{a, b, d\}], [\{d\}, \{a, b, c\}],$
 $[\{a, b\}, \{c, d\}], [\{a, c\}, \{b, d\}], [\{a, d\}, \{b, c\}]$

(3) $[\{a\}, \{b\}, \{c, d\}], [\{a\}, \{c\}, \{b, d\}], [\{a\}, \{d\}, \{b, c\}],$
 $[\{b\}, \{c\}, \{a, d\}], [\{b\}, \{d\}, \{a, c\}], [\{c\}, \{d\}, \{a, b\}]$

(4) $[\{a\}, \{b\}, \{c\}, \{d\}]$

There are 15 different partitions of S .

1.21 Let $\mathbf{N} = \{1, 2, 3, \dots\}$ and, for each $n \in \mathbf{N}$, Let $A_n = \{n, 2n, 3n, \dots\}$. Find:

(a) $A_3 \cap A_5$; (b) $A_4 \cap A_6$; (c) $\bigcap_{i \in Q} A_i$ where $Q = \{2, 3, 5, 7, 11, \dots\}$ is the set of prime numbers.

(a) Those numbers which are multiples of both 3 and 5 are the multiples of 15; hence $A_3 \cap A_5 = A_{15}$.

(b) The multiples of 12 and no other numbers belong to both A_4 and A_6 , hence $A_4 \cap A_6 = A_{12}$. (c)

Every positive integer except 1 is a multiple of at least one prime number; hence

$$\bigcap_{i \in Q} A_i = \{2, 3, 4, \dots\} = \mathbf{N} \setminus \{1\}$$

1.22 Let $\{A_i \mid i \in I\}$ be an indexed class of sets and let $i_0 \in I$. Prove

$$\bigcap_{i \in I} A_i \subseteq A_{i_0} \subseteq \bigcup_{i \in I} A_i$$

Let $x \in \bigcap_{i \in I} A_i$ then $x \in A_i$ for every $i \in I$. In particular, $x \in A_{i_0}$. Hence $\bigcap_{i \in I} A_i \subseteq A_{i_0}$. Now let $y \in A_{i_0}$. Since i_0

$\in I$, $y \in \bigcup_{i \in I} A_i$. Hence $A_{i_0} \subseteq \bigcup_{i \in I} A_i$.

1.23 Prove (De Morgan's law): For any indexed class $\{A_i \mid i \in I\}$, we have $\bigcap_i A_i^C = \bigcup_i A_i^C$.

Using the definitions of union and intersection of indexed classes of sets:

$$\begin{aligned}\bigcap_i A_i^C &= \{x \mid x \in A_i^C \text{ for every } i\} = \{x \mid x \notin A_i \text{ for every } i\} \\ &= \{x \mid x \in A_i^C \text{ for every } i\} = \bigcup_i A_i^C\end{aligned}$$

MATHEMATICAL INDUCTION

1.24 Prove the proposition $P(n)$ that the sum of the first n positive integers is $\frac{1}{2}n(n+1)$; that is,

$$P(n) = 1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1)$$

The proposition holds for $n = 1$ since:

$$P(1) : 1 = \frac{1}{2}(1)(1+1)$$

Assuming $P(k)$ is true, we add $k+1$ to both sides of $P(k)$, obtaining

$$\begin{aligned}1 + 2 + 3 + \cdots + k + (k+1) &= \frac{1}{2}k(k+1) + (k+1) \\ &= \frac{1}{2}[k(k+1) + 2(k+1)] \\ &= \frac{1}{2}[(k+1)(k+2)]\end{aligned}$$

which is $P(k+1)$. That is, $P(k+1)$ is true whenever $P(k)$ is true. By the Principle of Induction, P is true for all n .

1.25 Prove the following proposition (for $n \geq 0$):

$$P(n) : 1 + 2 + 2^2 + 2^3 + \cdots + 2^n = 2^{n+1} - 1$$

$P(0)$ is true since $1 = 2^1 - 1$. Assuming $P(k)$ is true, we add 2^{k+1} to both sides of $P(k)$, obtaining

$$1 + 2 + 2^2 + 2^3 + \cdots + 2^k + 2^{k+1} = 2^{k+1} - 1 + 2^{k+1} = 2(2^{k+1}) - 1 = 2^{k+2} - 1$$

which is $P(k+1)$. That is, $P(k+1)$ is true whenever $P(k)$ is true. By the principle of induction, $P(n)$ is true for all n .

Supplementary Problems

SETS AND SUBSETS

1.26 Which of the following sets are equal?

$$\begin{aligned}A &= \{x \mid x^2 - 4x + 3 = 0\}, \quad C = \{x \mid x \in \mathbb{N}, x < 3\}, \quad E = \{1, 2\}, \quad G = \{3, 1\}, \quad B = \{x \mid x^2 - 3x + 2 = 0\}, \\ D &= \{x \mid x \in \mathbb{N}, x \text{ is odd}, x < 5\}, \quad F = \{1, 2, 1\}, \quad H = \{1, 1, 3\}.\end{aligned}$$

1.27 List the elements of the following sets if the universal set is $U = \{a, b, c, \dots, y, z\}$.

Furthermore, identify which of the sets, if any, are equal.

$$\begin{aligned}A &= \{x \mid x \text{ is a vowel}\}, \quad C = \{x \mid x \text{ precedes } f \text{ in the alphabet}\}, \\ B &= \{x \mid x \text{ is a letter in the word "little"}\}, \quad D = \{x \mid x \text{ is a letter in the word "title"}\}.\end{aligned}$$

1.28 Let $A = \{1, 2, \dots, 8, 9\}$, $B = \{2, 4, 6, 8\}$, $C = \{1, 3, 5, 7, 9\}$, $D = \{3, 4, 5\}$, $E = \{3, 5\}$. Which of these sets can equal a set X under each of the following conditions?

- (a) X and B are disjoint. (c) $X \subseteq A$ but $X \subset C$.
(b) $X \subseteq D$ but $X \subset B$. (d) $X \subseteq C$ but $X \subset A$.

SET OPERATIONS

1.29 Consider the universal set $U = \{1, 2, 3, \dots, 8, 9\}$ and sets $A = \{1, 2, 5, 6\}$, $B = \{2, 5, 7\}$, $C = \{1, 3, 5, 7, 9\}$. Find: (a)

$A \cap B$ and $A \cap C$ (c) A^C and C^C (e) $A \oplus B$ and $A \oplus C$

(b) $A \cup B$ and $B \cup C$ (d) $A \setminus B$ and $A \setminus C$ (f) $(A \cup C) \setminus B$ and $(B \oplus C) \setminus A$

1.30 Let A and B be any sets. Prove:

(a) A is the disjoint union of $A \setminus B$ and $A \cap B$.

(b) $A \cup B$ is the disjoint union of $A \setminus B$, $A \cap B$, and $B \setminus A$.

1.31 Prove the following:

(a) $A \subseteq B$ if and only if $A \cap B^C = \emptyset$ (c) $A \subseteq B$ if and only if $B^C \subseteq A^C$

(b) $A \subseteq B$ if and only if $A^C \cup B = U$ (d) $A \subseteq B$ if and only if $A \setminus B = \emptyset$

(Compare the results with Theorem 1.4.)

1.32 Prove the Absorption Laws: (a) $A \cup (A \cap B) = A$; (b) $A \cap (A \cup B) = A$.

1.33 The formula $A \setminus B = A \cap B^C$ defines the difference operation in terms of the operations of intersection and complement. Find a formula that defines the union $A \cup B$ in terms of the operations of intersection and complement.

VENN DIAGRAMS

1.34 The Venn diagram in Fig. 1-5(a) shows sets A , B , C . Shade the following sets:

(a) $A \setminus (B \cup C)$; (b) $A^C \cap (B \cup C)$; (c) $A^C \cap (C \setminus B)$.

1.35 Use the Venn diagram in Fig. 1-5(b) to write each set as the (disjoint) union of fundamental products:

(a) $A \cap (B \cup C)$; (b) $A^C \cap (B \cup C)$; (c) $A \cup (B \setminus C)$.

1.36 Consider the following assumptions:

S_1 : All dictionaries are useful.

S_2 : Mary owns only romance novels.

S_3 : No romance novel is useful.

Use a Venn diagram to determine the validity of each of the following conclusions:

(a) Romance novels are not dictionaries.

(b) Mary does not own a dictionary.

(c) All useful books are dictionaries.

ALGEBRA OF SETS AND DUALITY

1.37 Write the dual of each equation:

(a) $A = (B^C \cap A) \cup (A \cap B)$

(b) $(A \cap B) \cup (A^C \cap B) \cup (A \cap B^C) \cup (A^C \cap B^C) = U$

1.38 Use the laws in Table 1-1 to prove each set identity:

(a) $(A \cap B) \cup (A \cap B^C) = A$

(b) $A \cup B = (A \cap B^C) \cup (A^C \cap B) \cup (A \cap B)$

FINITE SETS AND THE COUNTING PRINCIPLE

1.39 Determine which of the following sets are finite:

(a) Lines parallel to the x axis. (c) Integers which are multiples of 5.

(b) Letters in the English alphabet. (d) Animals living on the earth.

1.40 Use Theorem 1.9 to prove Corollary 1.10: Suppose A, B, C are finite sets. Then $A \cup B \cup C$ is finite and

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

1.41 A survey on a sample of 25 new cars being sold at a local auto dealer was conducted to see which of three popular options, air-conditioning (A), radio (R), and power windows (W), were already installed. The survey found:

15 had air-conditioning (A), 5 had A and P ,

12 had radio (R), 9 had A and R , 3 had all three options.

11 had power windows (W), 4 had R and W ,

Find the number of cars that had: (a) only W ; (b) only A ; (c) only R ; (d) R and W but not A ; (e) A and R but not W ; (f) only one of the options; (g) at least one option; (h) none of the options.

CLASSES OF SETS

1.42 Find the power set $P(A)$ of $A = \{1, 2, 3, 4, 5\}$.

1.43 Given $A = [\{a, b\}, \{c\}, \{d, e, f\}]$.

(a) List the elements of A . (b) Find $n(A)$. (c) Find the power set of A .

1.44 Suppose A is finite and $n(A) = m$. Prove the power set $P(A)$ has 2^m elements.

PARTITIONS

1.45 Let $S = \{1, 2, \dots, 8, 9\}$. Determine whether or not each of the following is a partition of S :

(a) $[\{1, 3, 6\}, \{2, 8\}, \{5, 7, 9\}]$ (c) $[\{2, 4, 5, 8\}, \{1, 9\}, \{3, 6, 7\}]$

(b) $[\{1, 5, 7\}, \{2, 4, 8, 9\}, \{3, 5, 6\}]$ (d) $[\{1, 2, 7\}, \{3, 5\}, \{4, 6, 8, 9\}, \{3, 5\}]$

1.46 Let $S = \{1, 2, 3, 4, 5, 6\}$. Determine whether or not each of the following is a partition of S :

(a) $P_1 = [\{1, 2, 3\}, \{1, 4, 5, 6\}]$ (c) $P_3 = [\{1, 3, 5\}, \{2, 4\}, \{6\}]$

(b) $P_2 = [\{1, 2\}, \{3, 5, 6\}]$ (d) $P_4 = [\{1, 3, 5\}, \{2, 4, 6, 7\}]$

1.47 Determine whether or not each of the following is a partition of the set $\mathbf{N}$ of positive integers:

(a) $[\{n \mid n > 5\}, \{n \mid n < 5\}]$; (b) $[\{n \mid n > 6\}, \{1, 3, 5\}, \{2, 4\}]$;

(c) $[\{n \mid n^2 > 11\}, \{n \mid n^2 < 11\}]$.

1.48 Let $[A_1, A_2, \dots, A_m]$ and $[B_1, B_2, \dots, B_n]$ be partitions of a set S .

Show that the following collection of sets is also a partition (called the *cross partition*) of S :

$$P = [A_i \cap B_j \mid i = 1, \dots, m, j = 1, \dots, n] \setminus \emptyset$$

Observe that we deleted the empty set $\emptyset$.

1.49 Let $S = \{1, 2, 3, \dots, 8, 9\}$. Find the cross partition P of the following partitions of S :

$$P_1 = [\{1, 3, 5, 7, 9\}, \{2, 4, 6, 8\}] \text{ and } P_2 = [\{1, 2, 3, 4\}, \{5, 7\}, \{6, 8, 9\}]$$

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INDUCTION

1.50 Prove: $2 + 4 + 6 + \dots + 2n = n(n + 1)$

1.51 Prove: $1 + 4 + 7 + \dots + 3n - 2 = \frac{n(3n-1)}{2}$

1.52 Prove: $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

1.53 Prove: $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$

$$\frac{(2n-1)(2n+1)}{2n+1} = \frac{n}{2n+1}$$

1.54 Prove: $\frac{1}{1 \cdot 5} + \frac{1}{5 \cdot 9} + \frac{1}{9 \cdot 13} + \cdots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$

$$\frac{(4n-3)(4n+1)}{4n+1} = \frac{n}{4n+1}$$

1.55 Prove $7^n - 2^n$ is divisible by 5 for all $n \in \mathbb{N}$

1.56 Prove $n^3 - 4n + 6$ is divisible by 3 for all $n \in \mathbb{N}$

1.57 Use the identity $1 + 2 + 3 + \cdots + n = n(n+1)/2$ to prove that

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + 3 + \cdots + n)^2$$

Miscellaneous Problems

1.58 Suppose $\mathbf{N} = \{1, 2, 3, \dots\}$ is the universal set, and

$$A = \{n \mid n \leq 6\}, B = \{n \mid 4 \leq n \leq 9\}, C = \{1, 3, 5, 7, 9\}, D = \{2, 3, 5, 7, 8\}.$$

Find: (a) $A \oplus B$; (b) $B \oplus C$; (c) $A \cap (B \oplus D)$; (d) $(A \cap B) \oplus (A \cap D)$.

1.59 Prove the following properties of the symmetric difference:

- (a) $(A \oplus B) \oplus C = A \oplus (B \oplus C)$ (Associative Law).
- (b) $A \oplus B = B \oplus A$ (Commutative Law).
- (c) If $A \oplus B = A \oplus C$, then $B = C$ (Cancellation Law).
- (d) $A \cap (B \oplus C) = (A \cap B) \oplus (A \cap C)$ (Distributive Law).

1.60 Consider m nonempty distinct sets $A_1, A_2, \dots, A_m$ in a universal set $\mathbf{U}$. Prove:

- (a) There are 2^m fundamental products of the m sets.
- (b) Any two fundamental products are disjoint.
- (c) $\mathbf{U}$ is the union of all the fundamental products.

1.34 See Fig. 1-10.

Answers to Supplementary

Problems 1.26 $B = C = E = F, A = D = G$

$= H$.

1.27 $A = \{a, e, i, o, u\}, B = D = \{l, i, t, e\}, C = \{a, b, c, d, e\}.$

1.28 (a) C and E ; (b) D and E ; (c) A, B , and D ; (d) None.

1.29 (a) $A \cap B = \{2, 5\}, A \cap C = \{1, 5\};$
 (b) $A \cup B = \{1, 2, 5, 6, 7\}, B \cup C = \{1, 2, 3, 5, 7, 9\};$ (c) $A^C = \{3, 4, 7, 8, 9\}, C^C = \{2, 4, 6, 8\};$
 (d) $A \setminus B = \{1, 6\}, A \setminus C = \{2, 6\};$
 (e) $A \oplus B = \{1, 6, 7\}, A \oplus C = \{2, 3, 6, 7, 9\};$ (f) $(A \cup C) \setminus B = \{1, 3, 6, 9\}, (B \oplus C) \setminus A = \{3, 9\}.$

1.33 $A \cup B = (A^C \cap B^C)^C.$

1.35 (a) $(A \cap B \cap C) \cup (A \cap B \cap C^C) \cup (A \cap B^C \cap C)$

(b) $(A^C \cap B \cap C^C) \cup (A^C \cap B \cap C) \cup (A^C \cap B^C \cap C)$

(c) $(A \cap B \cap C) \cup (A \cap B \cap C^C) \cup (A \cap B^C \cap C) \cup (A^C \cap B \cap C^C) \cup (A \cap B^C \cap C^C)$

1.36 The three premises yield the Venn diagram in Fig. 1-11(a). (a) and (b) are valid, but (c) is not valid.

1.37 (a) $A = (B^C \cup A) \cap (A \cup B)$

(b) $(A \cup B) \cap (A^C \cup B) \cap (A \cup B^C) \cap (A^C \cup B^C) = \emptyset$

1.39 (a) Infinite; (b) finite; (c) infinite; (d) finite.

Fig. 1-10

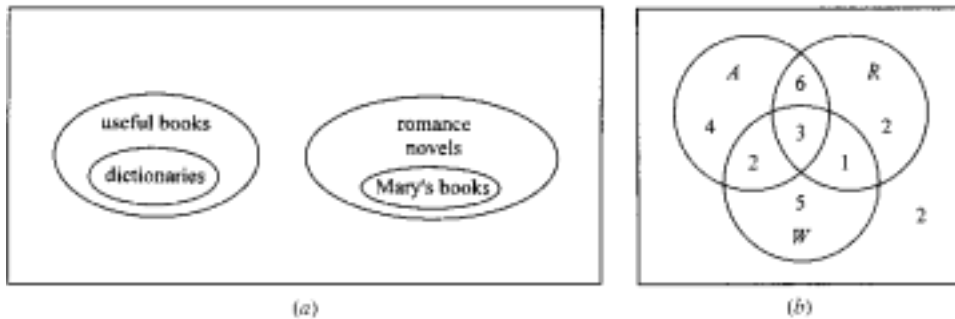


Fig. 1-11

1.41 Use the data to fill in the Venn diagram in Fig. 1-11(b). Then:

(a) 5; (b) 4; (c) 2; (d) 1; (e) 6; (f) 11; (g) 23; (h)

2. 1.42 $P(A)$ has $2^5 = 32$ elements as follows:

$\{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}, \{2, 3, 4\}, \{2, 3, 5\}, \{3, 4, 5\}, \{1, 3, 4\}, \{1, 3, 5\}, \{1, 4, 5\}, \{2, 4, 5\}, \{1, 2, 3, 4\}, \{1, 2, 3, 5\}, \{1, 2, 4, 5\}, \{1, 3, 4, 5\}, \{2, 3, 4, 5\}, A\}$

1.43 (a) Three elements: $\{a, b\}$, $\{c\}$, and $\{d, e, f\}$. (b) 3.

(c) $P(A)$ has $2^3 = 8$ elements as follows:

$P(A) = \{A, \{\{a, b\}, \{c\}\}, \{\{a, b\}, \{d, e, f\}\}, \{\{c\}, \{d, e, f\}\}, \{\{a, b\}\}, \{\{c\}\}, \{\{d, e, f\}\}, \emptyset\}$

1.44 Let X be an element in $P(A)$. For each $a \in A$, either $a \in X$ or $a \notin X$. Since $n(A) = m$, there are 2^m different sets X . That is $|P(A)| = 2^m$.

1.45 (a) No, (b) no, (c) yes, (d) yes.

1.46 (a) No, (b) no, (c) yes, (d) no.

1.47 (a) No, (b) no, (c) yes.

1.49 $\{\{1, 3\}, \{2, 4\}, \{5, 7\}, \{9\}, \{6, 8\}\}$

1.55 Hint: $7^{k+1} - 2^{k+1} = 7^{k+1} - 7(2^k) + 7(2^k) - 2^{k+1} = 7(7^k - 2^k) + (7 - 2)2^k$

1.58 (a) $\{1, 2, 3, 7, 8, 9\}$; (b) $\{1, 3, 4, 6, 8\}$; (c) and (d) $\{2, 3, 4, 6\}$.

Relations

2.1 INTRODUCTION

The reader is familiar with many relations such as “less than,” “is parallel to,” “is a subset of,” and so on. In a certain sense, these relations consider the existence or nonexistence of a certain connection between pairs of objects taken in a definite order. Formally, we define a relation in terms of these “ordered pairs.”

An *ordered pair* of elements a and b , where a is designated as the first element and b as the second element, is denoted by (a, b) . In particular,

$$(a, b) = (c, d)$$

if and only if $a = c$ and $b = d$. Thus $(a, b) = (b, a)$ unless $a = b$. This contrasts with sets where the order of elements is irrelevant; for example, $\{3, 5\} = \{5, 3\}$.

2.2 PRODUCT SETS

Consider two arbitrary sets A and B . The set of all ordered pairs (a, b) where $a \in A$ and $b \in B$ is called the *product*, or *Cartesian product*, of A and B . A short designation of this product is $A \times B$, which is read “ A cross B .” By definition,

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

One frequently writes A^2 instead of $A \times A$.

EXAMPLE 2.1 $\mathbf{R}$ denotes the set of real numbers and so $\mathbf{R}^2 = \mathbf{R} \times \mathbf{R}$ is the set of ordered pairs of real numbers. The reader is familiar with the geometrical representation of $\mathbf{R}^2$ as points in the plane as in Fig. 2-1. Here each point P represents an ordered pair (a, b) of real numbers and vice versa; the vertical line through P meets the x -axis at a , and the horizontal line through P meets the y -axis at b . $\mathbf{R}^2$ is frequently called the *Cartesian plane*.

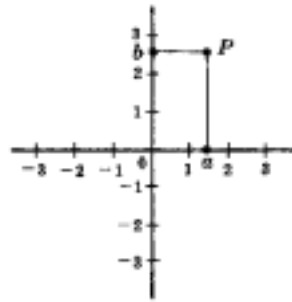
EXAMPLE 2.2 Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

$$B \times A = \{(a, 1), (b, 1), (c, 1), (a, 2), (b, 2), (c, 2)\}$$

Also, $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

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Fig. 2-1

There are two things worth noting in the above examples. First of all $A \times B = B \times A$. The Cartesian product deals with ordered pairs, so naturally the order in which the sets are considered is important. Secondly, using $n(S)$ for the number of elements in a set S , we have:

$$n(A \times B) = 6 = 2(3) = n(A)n(B)$$

In fact, $n(A \times B) = n(A)n(B)$ for any finite sets A and B . This follows from the observation that, for an ordered pair (a, b) in $A \times B$, there are $n(A)$ possibilities for a , and for each of these there are $n(B)$ possibilities for b .

The idea of a product of sets can be extended to any finite number of sets. For any sets $A_1, A_2, \dots, A_n$, the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ where $a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n$ is called the *product* of the sets $A_1, \dots, A_n$ and is denoted by

$$A_1 \times A_2 \times \cdots \times A_n$$

or $\prod_{i=1}^n A_i$

Just as we write A^2 instead of $A \times A$, so we write A^n instead of $A \times A \times \cdots \times A$, where there are n factors all equal to A . For example, $\mathbf{R}^3 = \mathbf{R} \times \mathbf{R} \times \mathbf{R}$ denotes the usual three-dimensional space.

2.3 RELATIONS

We begin with a definition.

Definition 2.1: Let A and B be sets. A *binary relation* or, simply, *relation* from A to B is a subset of $A \times B$.

Suppose R is a relation from A to B . Then R is a set of ordered pairs where each first element comes from A and each second element comes from B . That is, for each pair $a \in A$ and $b \in B$, exactly one of the following is true:

- (i) $(a, b) \in R$; we then say “ a is R -related to b ”, written aRb .
- (ii) $(a, b) \notin R$; we then say “ a is not R -related to b ”, written $a \not R b$.

If R is a relation from a set A to itself, that is, if R is a subset of $A^2 = A \times A$, then we say that R is a relation *on* A . The *domain* of a relation R is the set of all first elements of the ordered pairs which belong to R , and the *range* is the set of second elements.

Although n -ary relations, which involve ordered n -tuples, are introduced in Section 2.10, the term relation shall then mean binary relation unless otherwise stated or implied.

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EXAMPLE 2.3

- (a) $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$, and let $R = \{(1, y), (1, z), (3, y)\}$. Then R is a relation from A to B since R is a subset of $A \times B$. With respect to this relation,

$$1Ry, 1Rz, 3Ry, \text{ but } 1 \not R x, 2 \not R x, 2 \not R y, 2 \not R z, 3 \not R x, 3 \not R z$$

The domain of R is $\{1, 3\}$ and the range is $\{y, z\}$.

- (b) Set inclusion $\subseteq$ is a relation on any collection of sets. For, given any pair of set A and B , either $A \subseteq B$ or $A \not\subseteq B$.
- (c) A familiar relation on the set $\mathbf{Z}$ of integers is “ m divides n .” A common notation for this relation is to write $m|n$ when m divides n . Thus $6|30$ but $7 \nmid 25$.
- (d) Consider the set L of lines in the plane. Perpendicularity, written “ $\perp$,” is a relation on L . That is, given any pair of lines a and b , either $a \perp b$ or $a \not\perp b$. Similarly, “is parallel to,” written “ $\parallel$,” is a relation on L since either $a \parallel b$ or $a \not\parallel b$.
- (e) Let A be any set. An important relation on A is that of *equality*,

$$\{(a, a) | a \in A\}$$

which is usually denoted by “ $=$.” This relation is also called the *identity* or *diagonal* relation on A and it will also be denoted by Δ_A or simply Δ .

- (f) Let A be any set. Then $A \times A$ and $\emptyset$ are subsets of $A \times A$ and hence are relations on A called the *universal relation* and *empty relation*, respectively.

Inverse Relation

Let R be any relation from a set A to a set B . The *inverse* of R , denoted by R^{-1} , is the relation from B to A which consists of those ordered pairs which, when reversed, belong to R ; that is,

$$R^{-1} = \{(b, a) | (a, b) \in R\}$$

For example, let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. Then the inverse of

$$R = \{(1, y), (1, z), (3, y)\} \text{ is } R^{-1} = \{(y, 1), (z, 1), (y, 3)\}$$

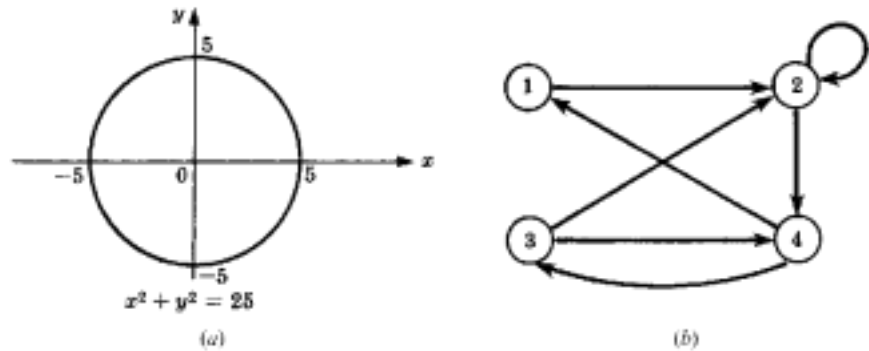
Clearly, if R is any relation, then $(R^{-1})^{-1} = R$. Also, the domain and range of R^{-1} are equal, respectively, to the range and domain of R . Moreover, if R is a relation on A , then R^{-1} is also a relation on A .

2.4 PICTORIAL REPRESENTATIVES OF RELATIONS

There are various ways of picturing relations.

Relations on $\mathbf{R}$

Let S be a relation on the set $\mathbf{R}$ of real numbers; that is, S is a subset of $\mathbf{R}^2 = \mathbf{R} \times \mathbf{R}$. Frequently, S consists of all ordered pairs of real numbers which satisfy some given equation $E(x, y) = 0$ (such as $x^2 + y^2 = 25$). Since $\mathbf{R}^2$ can be represented by the set of points in the plane, we can picture S by emphasizing those points in the plane which belong to S . The pictorial representation of the relation is sometimes called the *graph* of the relation. For example, the graph of the relation $x^2 + y^2 = 25$ is a circle having its center at the origin and radius 5. See Fig. 2-2(a).



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Fig. 2-2

Directed Graphs of Relations on Sets

There is an important way of picturing a relation R on a finite set. First we write down the elements of the set, and then we draw an arrow from each element x to each element y whenever x is related to y . This diagram is called the *directed graph* of the relation. Figure 2-2(b), for example, shows the directed graph of the following relation R on the set $A = \{1, 2, 3, 4\}$:

$$R = \{(1, 2), (2, 2), (2, 4), (3, 2), (3, 4), (4, 1), (4, 3)\}$$

Observe that there is an arrow from 2 to itself, since 2 is related to 2 under R .

These directed graphs will be studied in detail as a separate subject in Chapter 8. We mention it here mainly for completeness.

Pictures of Relations on Finite Sets

Suppose A and B are finite sets. There are two ways of picturing a relation R from A to B .

- (i) Form a rectangular array (matrix) whose rows are labeled by the elements of A and whose columns are labeled by the elements of B . Put a 1 or 0 in each position of the array according as $a \in A$ is or is not related to $b \in B$. This array is called the *matrix of the relation*.

- (ii) Write down the elements of A and the elements of B in two disjoint disks, and then draw an arrow from $a \in A$ to $b \in B$ whenever a is related to b . This picture will be called the *arrow diagram* of the relation.

Figure 2-3 pictures the relation R in Example 2.3(a) by the above two ways.

Fig. 2-3
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2.5 COMPOSITION OF RELATIONS

Let A , B and C be sets, and let R be a relation from A to B and let S be a relation from B to C . That is, R is a subset of $A \times B$ and S is a subset of $B \times C$. Then R and S give rise to a relation from A to C denoted by $R \circ S$ and defined by:

$$a(R \circ S)c \text{ if for some } b \in B \text{ we have } aRb \text{ and } bSc.$$

That is ,

$$R \circ S = \{(a, c) \mid \text{there exists } b \in B \text{ for which } (a, b) \in R \text{ and } (b, c) \in S\}$$

The relation $R \circ S$ is called the *composition* of R and S ; it is sometimes denoted simply by RS . Suppose R is a relation on a set A , that is, R is a relation from a set A to itself. Then $R \circ R$, the composition of R with itself, is always defined. Also, $R \circ R$ is sometimes denoted by R^2 . Similarly, $R^3 = R^2 \circ R = R \circ R \circ R$, and so on. Thus R^n is defined for all positive n .

Warning: Many texts denote the composition of relations R and S by $S \circ R$ rather than $R \circ S$. This is done in order to conform with the usual use of $g \circ f$ to denote the composition of f and g where f and g are functions. Thus the reader may have to adjust this notation when using this text as a supplement with another text. However, when a relation R is composed with itself, then the meaning of $R \circ R$ is unambiguous.

EXAMPLE 2.4 Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, $C = \{x, y, z\}$ and let

$$R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\} \text{ and } S = \{(b, x), (b, z), (c, y), (d, z)\}$$

Consider the arrow diagrams of R and S as in Fig. 2-4. Observe that there is an arrow from 2 to d which is followed by an arrow from d to z . We can view these two arrows as a “path” which “connects” the element $2 \in A$ to the element $z \in C$. Thus:

$$2(R \circ S)z \text{ since } 2Rd \text{ and } dSz$$

Similarly there is a path from 3 to x and a path from 3 to z . Hence

$$3(R \circ S)x \text{ and } 3(R \circ S)z$$

No other element of A is connected to an element of C . Accordingly,

$$R \circ S = \{(2, z), (3, x), (3, z)\}$$

Our first theorem tells us that composition of relations is associative.

Theorem 2.1: Let A , B , C and D be sets. Suppose R is a relation from A to B , S is a relation from B to C , and T is a relation from C to D . Then

$$(R \circ S) \circ T = R \circ (S \circ T)$$

We prove this theorem in Problem 2.8.

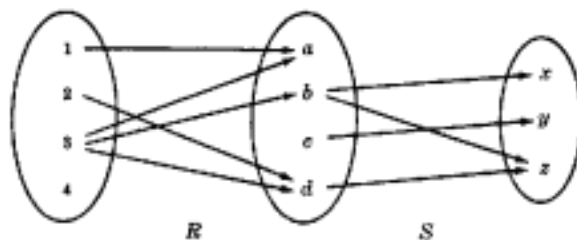


Fig. 2-4
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Composition of Relations and Matrices

There is another way of finding $R \circ S$. Let M_R and M_S denote respectively the matrix representations of the relations R and S . Then

$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix} \quad \text{and} \quad M_S = \begin{matrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Multiplying M_R and M_S we obtain the matrix

$$M = M_R M_S = \begin{matrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

The nonzero entries in this matrix tell us which elements are related by $R \circ S$. Thus $M = M_R M_S$ and $M_{R \circ S}$ have the same nonzero entries.

2.6 TYPES OF RELATIONS

This section discusses a number of important types of relations defined on a set A .

Reflexive Relations

A relation R on a set A is *reflexive* if aRa for every $a \in A$, that is, if $(a, a) \in R$ for every $a \in A$. Thus R is not reflexive if there exists $a \in A$ such that $(a, a) \notin R$.

EXAMPLE 2.5 Consider the following five relations on the set $A = \{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_3 = \{(1, 3), (2, 1)\}$$

$$R_4 = \emptyset, \text{ the empty relation}$$

$$R_5 = A \times A, \text{ the universal relation}$$

Determine which of the relations are reflexive.

Since A contains the four elements 1, 2, 3, and 4, a relation R on A is reflexive if it contains the four pairs $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. Thus only R_2 and the universal relation $R_5 = A \times A$ are reflexive. Note that R_1 , R_3 , and R_4 are not reflexive since, for example, $(2, 2)$ does not belong to any of them.

EXAMPLE 2.6 Consider the following five relations:

- (1) Relation $\leq$ (less than or equal) on the set $\mathbf{Z}$ of integers.
- (2) Set inclusion $\subseteq$ on a collection C of sets.
- (3) Relation $\perp$ (perpendicular) on the set L of lines in the plane.
- (4) Relation $\parallel$ (parallel) on the set L of lines in the plane.
- (5) Relation of divisibility on the set $\mathbf{N}$ of positive integers. (Recall $x \mid y$ if there exists z such that $xz = y$.)

Determine which of the relations are reflexive.

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The relation (3) is not reflexive since no line is perpendicular to itself. Also (4) is not reflexive since no line is parallel to itself. The other relations are reflexive; that is, $x \leq x$ for every $x \in \mathbf{Z}$, $A \subseteq A$ for any set $A \in C$, and $n \mid n$ for every positive integer $n \in \mathbf{N}$.

Symmetric and Antisymmetric Relations

A relation R on a set A is *symmetric* if whenever aRb then bRa , that is, if whenever $(a, b) \in R$ then $(b, a) \in R$. Thus R is not symmetric if there exists $a, b \in A$ such that $(a, b) \in R$ but $(b, a) \notin R$.

EXAMPLE 2.7

(a) Determine which of the relations in Example 2.5 are symmetric.

R_1 is not symmetric since $(1, 2) \in R_1$ but $(2, 1) \notin R_1$. R_3 is not symmetric since $(1, 3) \in R_3$ but $(3, 1) \notin R_3$. The other relations are symmetric.

(b) Determine which of the relations in Example 2.6 are symmetric.

The relation $\perp$ is symmetric since if line a is perpendicular to line b then b is perpendicular to a . Also, $\parallel$ is symmetric since if line a is parallel to line b then b is parallel to line a . The other relations are not symmetric. For example:

$$3 \leq 4 \text{ but } 4 \not\leq 3; \{1, 2\} \subseteq \{1, 2, 3\} \text{ but } \{1, 2, 3\} \not\subseteq \{1, 2\}; \text{ and } 2 \mid 6 \text{ but } 6 \not\mid 2.$$

A relation R on a set A is *antisymmetric* if whenever aRb and bRa then $a = b$, that is, if $a = b$ and aRb then bRa . Thus R is not antisymmetric if there exist distinct elements a and b in A such that aRb and bRa .

EXAMPLE 2.8

(a) Determine which of the relations in Example 2.5 are antisymmetric.

R_2 is not antisymmetric since $(1, 2)$ and $(2, 1)$ belong to R_2 , but $1 \neq 2$. Similarly, the universal relation R_3 is not antisymmetric. All the other relations are antisymmetric.

(b) Determine which of the relations in Example 2.6 are antisymmetric.

The relation $\leq$ is antisymmetric since whenever $a \leq b$ and $b \leq a$ then $a = b$. Set inclusion $\subseteq$ is antisymmetric since whenever $A \subseteq B$ and $B \subseteq A$ then $A = B$. Also, divisibility on $\mathbf{N}$ is antisymmetric since whenever $m \mid n$ and $n \mid m$ then $m = n$. (Note that divisibility on $\mathbf{Z}$ is not antisymmetric since $3 \mid -3$ and $-3 \mid 3$ but $3 \neq -3$.) The relations $\perp$ and $\parallel$ are not antisymmetric.

Remark: The properties of being symmetric and being antisymmetric are not negatives of each other. For

example, the relation $R = \{(1, 3), (3, 1), (2, 3)\}$ is neither symmetric nor antisymmetric. On the other hand, the relation $R = \{(1, 1), (2, 2)\}$ is both symmetric and antisymmetric.

Transitive Relations

A relation R on a set A is *transitive* if whenever aRb and bRc then aRc , that is, if whenever $(a, b), (b, c) \in R$ then $(a, c) \in R$. Thus R is not transitive if there exist $a, b, c \in R$ such that $(a, b), (b, c) \in R$ but $(a, c) \notin R$.

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EXAMPLE 2.9

(a) Determine which of the relations in Example 2.5 are transitive.

The relation R_3 is not transitive since $(2, 1), (1, 3) \in R_3$ but $(2, 3) \notin R_3$. All the other relations are transitive. (b) Determine which of the relations in Example 2.6 are transitive.

The relations $\leq$, $\subseteq$, and $\parallel$ are transitive, but certainly not $\perp$. Also, since no line is parallel to itself, we can have $a \parallel b$ and $b \parallel a$, but $a \not\parallel a$. Thus $\parallel$ is not transitive. (We note that the relation “is parallel or equal to” is a transitive relation on the set L of lines in the plane.)

The property of transitivity can also be expressed in terms of the composition of relations. For a relation R on A we did define $R^2 = R \circ R$ and, more generally, $R^n = R^{n-1} \circ R$. Then we have the following result:

Theorem 2.2: A relation R is transitive if and only if, for every $n \geq 1$, we have $R^n \subseteq R$.

2.7 CLOSURE PROPERTIES

Consider a given set A and the collection of all relations on A . Let P be a property of such relations, such as being symmetric or being transitive. A relation with property P will be called a P -relation. The P -closure of an arbitrary relation R on A , written $P(R)$, is a P -relation such that

$$R \subseteq P(R) \subseteq S$$

for every P -relation S containing R . We will write

$$\text{reflexive}(R), \text{symmetric}(R), \text{ and } \text{transitive}(R)$$

for the reflexive, symmetric, and transitive closures of R .

Generally speaking, $P(R)$ need not exist. However, there is a general situation where $P(R)$ will always exist. Suppose P is a property such that there is at least one P -relation containing R and that the intersection of any P -relations is again a P -relation. Then one can prove (Problem 2.16) that

$$P(R) = \bigcap \{S \mid S \text{ is a } P\text{-relation and } R \subseteq S\}$$

Thus one can obtain $P(R)$ from the “top-down,” that is, as the intersection of relations. However, one usually wants to find $P(R)$ from the “bottom-up,” that is, by adjoining elements to R to obtain $P(R)$. This we do below.

Reflexive and Symmetric Closures

The next theorem tells us how to obtain easily the reflexive and symmetric closures of a relation. Here $\Delta_A = \{(a, a) \mid a \in A\}$ is the diagonal or equality relation on A .

Theorem 2.3: Let R be a relation on a set A . Then:

- (i) $R \cup \Delta_A$ is the reflexive closure of R .
- (ii) $R \cup R^{-1}$ is the symmetric closure of R .

In other words, $\text{reflexive}(R)$ is obtained by simply adding to R those elements (a, a) in the diagonal which do not already belong to R , and $\text{symmetric}(R)$ is obtained by adding to R all pairs (b, a) whenever (a, b) belongs to R .

EXAMPLE 2.10 Consider the relation $R = \{(1, 1), (1, 3), (2, 4), (3, 1), (3, 3), (4, 3)\}$ on the set $A = \{1, 2, 3, 4\}$. Then

$$\text{reflexive}(R) = R \cup \{(2, 2), (4, 4)\} \text{ and } \text{symmetric}(R) = R \cup \{(4, 2), (3, 4)\}$$

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Transitive Closure

Let R be a relation on a set A . Recall that $R^2 = R \circ R$ and $R^n = R^{n-1} \circ R$. We define

R^j **Theorem 2.4:** R^* is the transitive closure of R .

The following theorem applies:

$$R^* = \bigcup_{i=1}^{\infty} R^i$$

Suppose A is a finite set with n elements. We show in Chapter 8 on graphs that

$$R^* = R \cup R^2 \cup \dots \cup R^n$$

This gives us the following theorem:

Theorem 2.5: Let R be a relation on a set A with n elements. Then

$$\text{transitive}(R) = R \cup R^2 \cup \dots \cup R^n$$

EXAMPLE 2.11 Consider the relation $R = \{(1, 2), (2, 3), (3, 3)\}$ on $A = \{1, 2, 3\}$. Then: $R^2 =$

$$R \circ R = \{(1, 3), (2, 3), (3, 3)\} \text{ and } R^3 = R^2 \circ R = \{(1, 3), (2, 3), (3, 3)\}$$

Accordingly,

$$\text{transitive}(R) = \{(1, 2), (2, 3), (3, 3), (1, 3)\}$$

2.8 EQUIVALENCE RELATIONS

Consider a nonempty set S . A relation R on S is an *equivalence relation* if R is reflexive, symmetric, and transitive. That is, R is an equivalence relation on S if it has the following three properties:

- (1) For every $a \in S$, aRa . (2) If aRb , then bRa . (3) If aRb and bRc , then aRc .

The general idea behind an equivalence relation is that it is a classification of objects which are in some way “alike.” In fact, the relation “=” of equality on any set S is an equivalence relation; that is:

- (1) $a = a$ for every $a \in S$. (2) If $a = b$, then $b = a$. (3) If $a = b$, $b = c$, then $a = c$. Other

equivalence relations follow.

EXAMPLE 2.12

(a) Let L be the set of lines and let T be the set of triangles in the Euclidean plane.

(i) The relation “is parallel to or identical to” is an equivalence relation on L .

(ii) The relations of congruence and similarity are equivalence relations on T .

- (b) The relation $\subseteq$ of set inclusion is not an equivalence relation. It is reflexive and transitive, but it is not symmetric since $A \subseteq B$ does not imply $B \subseteq A$.

- (c) Let m be a fixed positive integer. Two integers a and b are said to be *congruent modulo m* , written a

$$\equiv b \pmod{m}$$

if m divides $a - b$. For example, for the modulus $m = 4$, we have

$$11 \equiv 3 \pmod{4} \text{ and } 22 \equiv 6 \pmod{4}$$

since 4 divides $11 - 3 = 8$ and 4 divides $22 - 6 = 16$. This relation of congruence modulo m is an important equivalence relation.

Equivalence Relations and Partitions

This subsection explores the relationship between equivalence relations and partitions on a non-empty set S . Recall first that a partition P of S is a collection $\{A_i\}$ of nonempty subsets of S with the following two properties:

- (1) Each $a \in S$ belongs to some A_i .
- (2) If $A_i = A_j$ then $A_i \cap A_j = \emptyset$.

In other words, a partition P of S is a subdivision of S into disjoint nonempty sets. (See Section 1.7.)

Suppose R is an equivalence relation on a set S . For each $a \in S$, let $[a]$ denote the set of elements of S to which a is related under R ; that is:

$$[a] = \{x \mid (a, x) \in R\}$$

We call $[a]$ the *equivalence class* of a in S ; any $b \in [a]$ is called a *representative* of the equivalence class.

The collection of all equivalence classes of elements of S under an equivalence relation R is denoted by S/R , that is,

$$S/R = \{[a] \mid a \in S\}$$

It is called the *quotient set* of S by R . The fundamental property of a quotient set is contained in the following theorem.

Theorem 2.6: Let R be an equivalence relation on a set S . Then S/R is a partition of S . Specifically:

- (i) For each a in S , we have $a \in [a]$.
- (ii) $[a] = [b]$ if and only if $(a, b) \in R$.
- (iii) If $[a] = [b]$, then $[a]$ and $[b]$ are disjoint.

Conversely, given a partition $\{A_i\}$ of the set S , there is an equivalence relation R on S such that the sets A_i are the equivalence classes.

This important theorem will be proved in Problem 2.17.

EXAMPLE 2.13

- (a) Consider the relation $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$ on $S = \{1, 2, 3\}$.

One can show that R is reflexive, symmetric, and transitive, that is, that R is an equivalence relation. Also:

$$[1] = \{1, 2\}, [2] = \{1, 2\}, [3] = \{3\}$$

Observe that $[1] = [2]$ and that $S/R = \{[1], [3]\}$ is a partition of S . One can choose either $\{1, 3\}$ or $\{2, 3\}$ as a set of representatives of the equivalence classes.

(b) Let R_5 be the relation of congruence modulo 5 on the set $\mathbf{Z}$ of integers denoted by x

$$\equiv y \pmod{5}$$

This means that the difference $x - y$ is divisible by 5. Then R_5 is an equivalence relation on $\mathbf{Z}$. The quotient set $\mathbf{Z}/R_5$ contains the following five equivalence classes:

$$\begin{aligned} A_0 &= \{\dots, -10, -5, 0, 5, 10, \dots\} \\ A_1 &= \{\dots, -9, -4, 1, 6, 11, \dots\} \\ A_2 &= \{\dots, -8, -3, 2, 7, 12, \dots\} \\ A_3 &= \{\dots, -7, -2, 3, 8, 13, \dots\} \\ A_4 &= \{\dots, -6, -1, 4, 9, 14, \dots\} \end{aligned}$$

Any integer x , uniquely expressed in the form $x = 5q + r$ where $0 \leq r < 5$, is a member of the equivalence class A_r , where r is the remainder. As expected, $\mathbf{Z}$ is the disjoint union of equivalence classes A_1, A_2, A_3, A_4 . Usually one chooses $\{0, 1, 2, 3, 4\}$ or $\{-2, -1, 0, 1, 2\}$ as a set of representatives of the equivalence classes.

2.9 PARTIAL ORDERING RELATIONS

A relation R on a set S is called a *partial ordering* or a *partial order* of S if R is reflexive, antisymmetric, and transitive. A set S together with a partial ordering R is called a *partially ordered set* or *poset*. Partially ordered sets will be studied in more detail in Chapter 14, so here we simply give some examples.

EXAMPLE 2.14

(a) The relation $\subseteq$ of set inclusion is a partial ordering on any collection of sets since set inclusion has the three desired properties. That is,

- (1) $A \subseteq A$ for any set A .
- (2) If $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- (3) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

(b) The relation $\leq$ on the set $\mathbf{R}$ of real numbers is reflexive, antisymmetric, and transitive. Thus $\leq$ is a partial ordering on $\mathbf{R}$.

(c) The relation “ a divides b ,” written $a \mid b$, is a partial ordering on the set $\mathbf{N}$ of positive integers. However, “ a divides b ” is not a partial ordering on the set $\mathbf{Z}$ of integers since $a \mid b$ and $b \mid a$ need not imply $a = b$. For example, $3 \mid -3$ and $-3 \mid 3$ but $3 \neq -3$.

2.10 n -ARY RELATIONS

All the relations discussed above were binary relations. By an *n -ary relation*, we mean a set of ordered n -tuples. For any set S , a subset of the product set S^n is called an *n -ary relation* on S . In particular, a subset of S^3 is called a *ternary relation* on S .

EXAMPLE 2.15

(a) Let L be a line in the plane. Then “betweenness” is a ternary relation R on the points of L ; that is, $(a, b, c) \in R$ if b lies between a and c on L .

(b) The equation $x^2 + y^2 + z^2 = 1$ determines a ternary relation T on the set $\mathbf{R}$ of real numbers. That is, a triple (x, y, z) belongs to T if (x, y, z) satisfies the equation, which means (x, y, z) is the coordinates of a point in $\mathbf{R}^3$ on the sphere S with radius 1 and center at the origin $O = (0, 0, 0)$.

PRODUCT SETS

2.1. Given: $A = \{1, 2\}$, $B = \{x, y, z\}$, and $C = \{3, 4\}$. Find: $A \times B \times C$.

$A \times B \times C$ consists of all ordered triplets (a, b, c) where $a \in A$, $b \in B$, $c \in C$. These elements of $A \times B \times C$ can be systematically obtained by a so-called tree diagram (Fig. 2-5). The elements of $A \times B \times C$ are precisely the 12 ordered triplets to the right of the tree diagram.

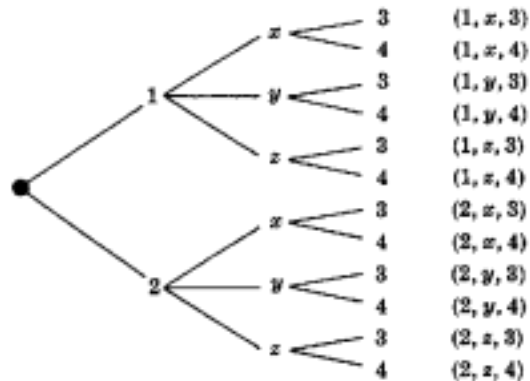


Fig. 2-5

Observe that $n(A) = 2$, $n(B) = 3$, and $n(C) = 2$ and, as expected,

$$n(A \times B \times C) = 12 = n(A) \cdot n(B) \cdot n(C)$$

2.2. Find x and y given $(2x, x + y) = (6, 2)$.

Two ordered pairs are equal if and only if the corresponding components are equal. Hence we obtain the equations

$$2x = 6 \text{ and } x + y = 2$$

from which we derive the answers $x = 3$ and $y = -1$.

RELATIONS AND THEIR GRAPHS

2.3. Find the number of relations from $A = \{a, b, c\}$ to $B = \{1, 2\}$.

There are $3(2) = 6$ elements in $A \times B$, and hence there are $m = 2^6 = 64$ subsets of $A \times B$. Thus there are $m = 64$ relations from A to B .

2.4. Given $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$. Let R be the following relation from A to B : R

$$= \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$$

(a) Determine the matrix of the relation.

(b) Draw the arrow diagram of R .

(c) Find the inverse relation R^{-1} of R .

(d) Determine the domain and range of R .

(a) See Fig. 2-6(a) Observe that the rows of the matrix are labeled by the elements of A and the columns by the elements of B . Also observe that the entry in the matrix corresponding to $a \in A$ and $b \in B$ is 1 if a is related to b and 0 otherwise.

(b) See Fig. 2.6(b) Observe that there is an arrow from $a \in A$ to $b \in B$ iff a is related to b , i.e., iff $(a, b) \in R$.

Fig. 2-6

(c) Reverse the ordered pairs of R to obtain R^{-1} :

$$R^{-1} = \{(y, 1), (z, 1), (y, 3), (x, 4), (z, 4)\}$$

Observe that by reversing the arrows in Fig. 2.6(b), we obtain the arrow diagram of R^{-1} .

(d) The domain of R , $\text{Dom}(R)$, consists of the first elements of the ordered pairs of R , and the range of R , $\text{Ran}(R)$, consists of the second elements. Thus,

$$\text{Dom}(R) = \{1, 3, 4\} \text{ and } \text{Ran}(R) = \{x, y, z\}$$

2.5. Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, and $C = \{x, y, z\}$. Consider the following relations R and S from A to B and from B to C , respectively.

$$R = \{(1, b), (2, a), (2, c)\} \text{ and } S = \{(a, y), (b, x), (c, y), (c, z)\}$$

(a) Find the composition relation $R \circ S$.

(b) Find the matrices M_R , M_S , and $M_{R \circ S}$ of the respective relations R , S , and $R \circ S$, and compare $M_{R \circ S}$ to the product $M_R M_S$.

(a) Draw the arrow diagram of the relations R and S as in Fig. 2-7(a). Observe that 1 in A is “connected” to x in C by the path $1 \rightarrow b \rightarrow x$; hence $(1, x)$ belongs to $R \circ S$. Similarly, $(2, y)$ and $(2, z)$ belong to $R \circ S$. We have

$$R \circ S = \{(1, x), (2, y), (2, z)\}$$

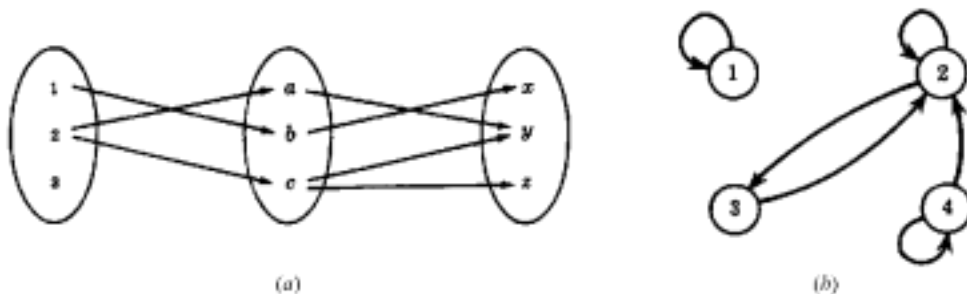


Fig. 2-7

36 RELATIONS [CHAP. 2 (b) The matrices of M_R , M_S , and $M_{R \circ S}$ follow:

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad M_S = \begin{matrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix} \quad M_{R \circ S} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} x & y & z \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$\square$ 100 021 000
 Multiplying M_R and $\square$
 M_S we obtain $M_R M_S =$

Observe that $M_{R \circ S}$ and $M_R M_S$ have the same zero entries.

2.6. Consider the relation $R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}$ on $A = \{1, 2, 3, 4\}$.

(a) Draw its directed graph. (b) Find $R^2 = R \circ R$.

(a) For each $(a, b) \in R$, draw an arrow from a to b as in Fig. 2-7(b).

(b) For each pair $(a, b) \in R$, find all $(b, c) \in R$. Then $(a, c) \in R^2$. Thus

$$R^2 = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (4, 3), (4, 4)\}$$

2.7. Let R and S be the following relations on $A = \{1, 2, 3\}$:

$$R = \{(1, 1), (1, 2), (2, 3), (3, 1), (3, 3)\}, S = \{(1, 2), (1, 3), (2, 1), (3, 3)\}$$

Find (a) $R \cup S$, $R \cap S$, R^C ; (b) $R \circ S$; (c) $S^2 = S \circ S$.

(a) Treat R and S simply as sets, and take the usual intersection and union. For R^C , use the fact that $A \times A$ is the universal relation on A .

$$\begin{aligned} R \cap S &= \{(1, 2), (3, 3)\} \\ R \cup S &= \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 3)\} \\ R^C &= \{(1, 3), (2, 1), (2, 2), (3, 2)\} \end{aligned}$$

(b) For each pair $(a, b) \in R$, find all pairs $(b, c) \in S$. Then $(a, c) \in R \circ S$. For example, $(1, 1) \in R$ and $(1, 2), (1, 3) \in S$; hence $(1, 2)$ and $(1, 3)$ belong to $R \circ S$. Thus,

$$R \circ S = \{(1, 2), (1, 3), (1, 1), (2, 3), (3, 2), (3, 3)\}$$

(c) Following the algorithm in (b), we get

$$S^2 = S \circ S = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 3)\}$$

2.8. Prove Theorem 2.1: Let A, B, C and D be sets. Suppose R is a relation from A to B , S is a relation from B to C and T is a relation from C to D . Then $(R \circ S) \circ T = R \circ (S \circ T)$.

We need to show that each ordered pair in $(R \circ S) \circ T$ belongs to $R \circ (S \circ T)$, and vice versa.

Suppose (a, d) belongs to $(R \circ S) \circ T$. Then there exists $c \in C$ such that $(a, c) \in R \circ S$ and $(c, d) \in T$. Since $(a, c) \in R \circ S$, there exists $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. Since $(b, c) \in S$ and $(c, d) \in T$, we have $(b, d) \in S \circ T$; and since $(a, b) \in R$ and $(b, d) \in S \circ T$, we have $(a, d) \in R \circ (S \circ T)$. Therefore, $(R \circ S) \circ T \subseteq R \circ (S \circ T)$. Similarly $R \circ (S \circ T) \subseteq (R \circ S) \circ T$. Both inclusion relations prove $(R \circ S) \circ T = R \circ (S \circ T)$.

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TYPES OF RELATIONS AND CLOSURE PROPERTIES

2.9. Consider the following five relations on the set $A = \{1, 2, 3\}$:

$$\begin{aligned} R &= \{(1, 1), (1, 2), (1, 3), (3, 3)\}, \varnothing = \text{empty relation} \\ S &= \{(1, 1)(1, 2), (2, 1)(2, 2), (3, 3)\}, A \times A = \text{universal relation} \\ T &= \{(1, 1), (1, 2), (2, 2), (2, 3)\} \end{aligned}$$

Determine whether or not each of the above relations on A is: (a) reflexive; (b) symmetric; (c) transitive; (d) antisymmetric.

- (a) R is not reflexive since $2 \in A$ but $(2, 2) \notin R$. T is not reflexive since $(3, 3) \notin T$ and, similarly, $\varnothing$ is not reflexive. S and $A \times A$ are reflexive.
- (b) R is not symmetric since $(1, 2) \in R$ but $(2, 1) \notin R$, and similarly T is not symmetric. S , $\varnothing$, and $A \times A$ are symmetric.
- (c) T is not transitive since $(1, 2)$ and $(2, 3)$ belong to T , but $(1, 3)$ does not belong to T . The other four relations are transitive.
- (d) S is not antisymmetric since $1 = 2$, and $(1, 2)$ and $(2, 1)$ both belong to S . Similarly, $A \times A$ is not antisymmetric. The other three relations are antisymmetric.

2.10. Give an example of a relation R on $A = \{1, 2, 3\}$ such that:

- (a) R is both symmetric and antisymmetric.
- (b) R is neither symmetric nor antisymmetric.
- (c) R is transitive but $R \cup R^{-1}$ is not transitive.

There are several such examples. One possible set of examples follows:

(a) $R = \{(1, 1), (2, 2)\}$; (b) $R = \{(1, 2), (2, 3)\}$; (c) $R = \{(1, 2)\}$.

2.11. Suppose C is a collection of relations S on a set A , and let T be the intersection of the relations S in C , that is, $T = \bigcap \{S \mid S \in C\}$. Prove:

- (a) If every S is symmetric, then T is symmetric.
- (b) If every S is transitive, then T is transitive.
- (a) Suppose $(a, b) \in T$. Then $(a, b) \in S$ for every S . Since each S is symmetric, $(b, a) \in S$ for every S . Hence $(b, a) \in T$ and T is symmetric.
- (b) Suppose (a, b) and (b, c) belong to T . Then (a, b) and (b, c) belong to S for every S . Since each S is transitive, (a, c) belongs to S for every S . Hence, $(a, c) \in T$ and T is transitive.

2.12. Let R be a relation on a set A , and let P be a property of relations, such as symmetry and transitivity. Then P will be called R -closable if P satisfies the following two conditions:

- (1) There is a P -relation S containing R .
- (2) The intersection of P -relations is a P -relation.
- (a) Show that symmetry and transitivity are R -closable for any relation R .
- (b) Suppose P is R -closable. Then $P(R)$, the P -closure of R , is the intersection of all P -relations S containing R , that is,

$$P(R) = \bigcap \{S \mid S \text{ is a } P\text{-relation and } R \subseteq S\}$$

- (a) The universal relation $A \times A$ is symmetric and transitive and $A \times A$ contains any relation R on A . Thus (1) is satisfied. By Problem 2.11, symmetry and transitivity satisfy (2). Thus symmetry and transitivity are R -closable for any relation R .

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- (b) Let $T = \bigcap \{S \mid S \text{ is a } P\text{-relation and } R \subseteq S\}$. Since P is R -closable, T is nonempty by (1) and T is a P -relation by (2). Since each relation S contains R , the intersection T contains R . Thus, T is a P -relation containing R . By definition, $P(R)$ is the smallest P -relation containing R ; hence $P(R) \subseteq T$. On the other hand, $P(R)$ is one of the sets S defining T , that is, $P(R)$ is a P -relation and if $R \subseteq P(R)$. Therefore, $T \subseteq P(R)$. Accordingly, $P(R) = T$.

2.13. Consider the relation $R = \{(a, a), (a, b), (b, c), (c, c)\}$ on the set $A = \{a, b, c\}$. Find: (a) reflexive(R); (b) symmetric(R); (c) transitive(R).

- (a) The reflexive closure on R is obtained by adding all diagonal pairs of $A \times A$ to R which are not currently in R . Hence,

$$\text{reflexive}(R) = R \cup \{(b, b)\} = \{(a, a), (a, b), (b, b), (b, c), (c, c)\}$$

- (b) The symmetric closure on R is obtained by adding all the pairs in R^{-1} to R which are not currently in R . Hence,

$$\text{symmetric}(R) = R \cup \{(b, a), (c, b)\} = \{(a, a), (a, b), (b, a), (b, c), (c, b), (c, c)\}$$

- (c) The transitive closure on R , since A has three elements, is obtained by taking the union of R with $R^2 = R \circ R$ and $R^3 = R \circ R \circ R$. Note that

$$R^2 = R \circ R = \{(a, a), (a, b), (a, c), (b, c), (c, c)\}$$

$$R^3 = R \circ R \circ R = \{(a, a), (a, b), (a, c), (b, c), (c, c), (a, b), (a, c), (b, c), (c, c)\}$$

Hence

$$\text{transitive}(R) = R \cup R^2 \cup R^3 = \{(a, a),$$

EQUIVALENCE RELATIONS AND PARTITIONS

2.14. Consider the $\mathbf{Z}$ of integers and an integer $m > 1$. We say that x is congruent to y modulo m , written

$$x \equiv y \pmod{m}$$

if $x - y$ is divisible by m . Show that this defines an equivalence relation on $\mathbf{Z}$.

We must show that the relation is reflexive, symmetric, and transitive.

- (i) For any x in $\mathbf{Z}$ we have $x \equiv x \pmod{m}$ because $x - x = 0$ is divisible by m . Hence the relation is reflexive.
- (ii) Suppose $x \equiv y \pmod{m}$, so $x - y$ is divisible by m . Then $-(x - y) = y - x$ is also divisible by m , so $y \equiv x \pmod{m}$. Thus the relation is symmetric.
- (iii) Now suppose $x \equiv y \pmod{m}$ and $y \equiv z \pmod{m}$, so $x - y$ and $y - z$ are each divisible by m . Then the sum $(x - y) + (y - z) = x - z$

$$- y) + (y - z) = x - z$$

is also divisible by m ; hence $x \equiv z \pmod{m}$. Thus the relation is transitive.

Accordingly, the relation of congruence modulo m on $\mathbf{Z}$ is an equivalence relation.

2.15. Let A be a set of nonzero integers and let $\approx$ be the relation on $A \times A$ defined by

$$(a, b) \approx (c, d) \text{ whenever } ad = bc$$

Prove that $\approx$ is an equivalence relation.

We must show that $\approx$ is reflexive, symmetric, and transitive.

- (i) *Reflexivity*: We have $(a, b) \approx (a, b)$ since $ab = ba$. Hence $\approx$ is reflexive.
- (ii) *Symmetry*: Suppose $(a, b) \approx (c, d)$. Then $ad = bc$. Accordingly, $cb = da$ and hence $(c, d) \approx (a, b)$. Thus, $\approx$ is symmetric.
- (iii) *Transitivity*: Suppose $(a, b) \approx (c, d)$ and $(c, d) \approx (e, f)$. Then $ad = bc$ and $cf = de$. Multiplying corresponding terms of the equations gives $(ad)(cf) = (bc)(de)$. Canceling $c \neq 0$ and $d \neq 0$ from both sides of the equation yields $af = be$, and hence $(a, b) \approx (e, f)$. Thus $\approx$ is transitive. Accordingly, $\approx$ is an equivalence relation.

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2.16. Let R be the following equivalence relation on the set $A = \{1, 2, 3, 4, 5, 6\}$:

$$R = \{(1, 1), (1, 5), (2, 2), (2, 3), (2, 6), (3, 2), (3, 3), (3, 6), (4, 4), (5, 1), (5, 5), (6, 2), (6, 3), (6, 6)\}$$

Find the partition of A induced by R , i.e., find the equivalence classes of R .

Those elements related to 1 are 1 and 5 hence

$$[1] = \{1, 5\}$$

We pick an element which does not belong to $[1]$, say 2. Those elements related to 2 are 2, 3, and 6, hence

$$[2] = \{2, 3, 6\}$$

The only element which does not belong to $[1]$ or $[2]$ is 4. The only element related to 4 is 4. Thus

$$[4] = \{4\}$$

Accordingly, the following is the partition of A induced by R :

$$\{\{1, 5\}, \{2, 3, 6\}, \{4\}\}$$

2.17. Prove Theorem 2.6: Let R be an equivalence relation in a set A . Then the quotient set A/R is a partition of A . Specifically,

- (i) $a \in [a]$, for every $a \in A$.
- (ii) $[a]=[b]$ if and only if $(a, b) \in R$.
- (iii) If $[a] = [b]$, then $[a]$ and $[b]$ are disjoint.
- (a) *Proof of (i):* Since R is reflexive, $(a, a) \in R$ for every $a \in A$ and therefore $a \in [a]$.
- (b) *Proof of (ii):* Suppose $(a, b) \in R$. We want to show that $[a]=[b]$. Let $x \in [b]$; then $(b, x) \in R$. But by hypothesis $(a, a) \in R$ and so, by transitivity, $(a, x) \in R$. Accordingly $x \in [a]$. Thus $[b] \subseteq [a]$. To prove that $[a] \subseteq [b]$ we observe that $(a, b) \in R$ implies, by symmetry, that $(b, a) \in R$. Then, by a similar argument, we obtain $[a] \subseteq [b]$. Consequently, $[a]=[b]$.
- On the other hand, if $[a]=[b]$, then, by (i), $b \in [b]=[a]$; hence $(a, b) \in R$.
- (c) *Proof of (iii):* We prove the equivalent contrapositive statement:

$$\text{If } [a] \cap [b] = \emptyset \text{ then } [a] \neq [b]$$

If $[a] \cap [b] = \emptyset$, then there exists an element $x \in A$ with $x \in [a] \cap [b]$. Hence $(a, x) \in R$ and $(b, x) \in R$. By symmetry, $(x, b) \in R$ and by transitivity, $(a, b) \in R$. Consequently by (ii), $[a]=[b]$.

PARTIAL ORDERINGS

2.18. Let $\mathcal{A}$ be any collection of sets. Is the relation of set inclusion $\subseteq$ a partial order on $\mathcal{A}$?

Yes, since set inclusion is reflexive, antisymmetric, and transitive. That is, for any sets A, B, C in $\mathcal{A}$ we have: (i) $A \subseteq A$; (ii) if $A \subseteq B$ and $B \subseteq A$, then $A = B$; (iii) if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

2.19. Consider the set $\mathbf{Z}$ of integers. Define aRb by $b = a^r$ for some positive integer r . Show that R is a partial order on $\mathbf{Z}$, that is, show that R is: (a) reflexive; (b) antisymmetric; (c) transitive.

- (a) R is reflexive since $a = a^1$.
- (b) Suppose aRb and bRa , say $b = a^r$ and $a = b^s$. Then $a = (a^r)^s = a^{rs}$. There are three possibilities: (i) $rs = 1$, (ii) $a = 1$, and (iii) $a = -1$. If $rs = 1$ then $r = 1$ and $s = 1$ and so $a = b$. If $a = 1$ then $b = 1^r = 1 = a$, and, similarly, if $b = 1$ then $a = 1$. Lastly, if $a = -1$ then $b = -1$ (since $b = 1$) and $a = b$. In all three cases, $a = b$. Thus R is antisymmetric.
- (c) Suppose aRb and bRc say $b = a^r$ and $c = b^s$. Then $c = (a^r)^s = a^{rs}$ and, therefore, aRc . Hence R is transitive.

Accordingly, R is a partial order on $\mathbf{Z}$.

40 RELATIONS [CHAP. 2] Supplementary Problems

RELATIONS

2.20. Let $S = \{a, b, c\}$, $T = \{b, c, d\}$, and $W = \{a, d\}$. Find $S \times T \times W$.

2.21. Find x and y where: (a) $(x + 2, 4) = (5, 2x + y)$; (b) $(y - 2, 2x + 1) = (x - 1, y + 2)$. **2.22.**

Prove: (a) $A \times (B \cap C) = (A \times B) \cap (A \times C)$; (b) $A \times (B \cup C) = (A \times B) \cup (A \times C)$. **2.23.**

Consider the relation $R = \{(1, 3), (1, 4), (3, 2), (3, 3), (3, 4)\}$ on $A = \{1, 2, 3, 4\}$.

- (a) Find the matrix M_R of R . (d) Draw the directed graph of R .
- (b) Find the domain and range of R . (e) Find the composition relation $R \circ R$.
- (c) Find R^{-1} . (f) Find $R \circ R^{-1}$ and $R^{-1} \circ R$.

2.24. Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $C = \{x, y, z\}$. Consider the relations R from A to B and S from B to C as follows: R

$$= \{(1, b), (3, a), (3, b), (4, c)\} \text{ and } S = \{(a, y), (c, x), (a, z)\}$$

- (a) Draw the diagrams of R and S .
- (b) Find the matrix of each relation R, S (composition) $R \circ S$.
- (c) Write R^{-1} and the composition $R \circ S$ as sets of ordered pairs.

2.25. Let R and S be the following relations on $B = \{a, b, c, d\}$:

$$R = \{(a, a), (a, c), (c, b), (c, d), (d, b)\} \text{ and } S = \{(b, a), (c, c), (c, d), (d, a)\}$$

Find the following composition relations: (a) $R \circ S$; (b) $S \circ R$; (c) $R \circ R$; (d) $S \circ S$.

2.26. Let R be the relation on $\mathbf{N}$ defined by $x + 3y = 12$, i.e. $R = \{(x, y) \mid x + 3y = 12\}$.

- (a) Write R as a set of ordered pairs. (c) Find R^{-1} .
 (b) Find the domain and range of R . (d) Find the composition relation $R \circ R$.

PROPERTIES OF RELATIONS

2.27. Each of the following defines a relation on the positive integers $\mathbf{N}$:

- (1) “ x is greater than y .” (3) $x + y = 10$
 (2) “ xy is the square of an integer.” (4) $x + 4y = 10$.

Determine which of the relations are: (a) reflexive; (b) symmetric; (c) antisymmetric; (d) transitive.

2.28. Let R and S be relations on a set A . Assuming A has at least three elements, state whether each of the following statements is true or false. If it is false, give a counterexample on the set $A = \{1, 2, 3\}$:

- (a) If R and S are symmetric then $R \cap S$ is symmetric.
 (b) If R and S are symmetric then $R \cup S$ is symmetric.
 (c) If R and S are reflexive then $R \cap S$ is reflexive.

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- (d) If R and S are reflexive then $R \cup S$ is reflexive.
 (e) If R and S are transitive then $R \cup S$ is transitive.
 (f) If R and S are antisymmetric then $R \cup S$ is antisymmetric.
 (g) If R is antisymmetric, then R^{-1} is antisymmetric.
 (h) If R is reflexive then $R \cap R^{-1}$ is not empty.
 (i) If R is symmetric then $R \cap R^{-1}$ is not empty.

2.29. Suppose R and S are relations on a set A , and R is antisymmetric. Prove that $R \cap S$ is antisymmetric.

EQUIVALENCE RELATIONS

2.30. Prove that if R is an equivalence relation on a set A , then R^{-1} is also an equivalence relation on A .

2.31. Let $S = \{1, 2, 3, \dots, 18, 19\}$. Let R be the relation on S defined by “ xy is a square,” (a) Prove R is an equivalence relation. (b) Find the equivalence class $[1]$. (c) List all equivalence classes with more than one element.

2.32. Let $S = \{1, 2, 3, \dots, 14, 15\}$. Let R be the equivalence relation on S defined by $x \equiv y \pmod{5}$, that is, $x - y$ is divisible by 5. Find the partition of S induced by R , i.e. the quotient set S/R .

2.33. Let $S = \{1, 2, 3, \dots, 9\}$, and let $\sim$ be the relation on $A \times A$ defined by

$$(a, b) \sim (c, d) \text{ whenever } a + d = b + c.$$

- (a) Prove that $\sim$ is an equivalence relation.
 (b) Find $[(2, 5)]$, that is, the equivalence class of $(2, 5)$.

2.21. (a) $x = 3, y = -2$; (b) $x = 2, y = 3$.

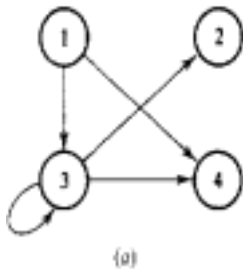
2.23.

(a) $M_R = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$;

(b) Domain = $\{1, 3\}$, range = $\{2, 3, 4\}$; (c)
 $R^{-1} = \{(3, 1), (4, 1), (2, 3), (3, 3), (4, 3)\}$;

Answers to Supplementary Problems

2.20. $\{(a, b, a), (a, b, d), (a, c, a), (a, c, d), (a, d, a), (a, d, d), (b, b, a), (b, b, d), (b, c, a), (b, c, d), (b, d, a), (b, d, d), (c, b, a), (c, b, d), (c, c, a), (c, c, d), (c, d, a), (c, d, d)\}$



- 2.25. (a) $R \circ S = \{(a, c), (a, d), (c, a), (d, a)\}$
 (b) $S \circ R = \{(b, a), (b, c), (c, b), (c, d), (d, a), (d, c)\}$
 (c) $R \circ R = \{(a, a), (a, b), (a, c), (a, d), (c, b)\}$
 (d) $S \circ S = \{(c, c), (c, a), (c, d)\}$

(d) See Fig. 2-8(a);

- (e) $R \circ R = \{(1, 2), (1, 3), (1, 4), (3, 2), (3, 3), (3, 4)\}$.

2.24. (a) See Fig. 2-8(b);

- (b) $R = [0, 1, 0; 0, 0, 0; 1, 1, 0; 0, 0, 1]$,
 $S = [0, 1, 1; 0, 0, 0; 1, 0, 0]$,
 $R \circ S = [0, 0, 0; 0, 0, 0; 0, 1, 1; 1, 0, 0]$; **Fig. 2-8**
 (c) $\{(b, 1), (a, 3), (b, 3), (c, 4)\}, \{(3, y), (3, z), (4, x)\}$.

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$$R = \{(1, 2)\}, S = \{(2, 1)\}.$$

- 2.31. (b) $\{1, 4, 9, 16\}$; (c) $\{1, 4, 9, 16\}, \{2, 8, 18\}, \{3, 12\}$.

- 2.26. (a) $\{(9, 1), (6, 2), (3, 3)\}$; (b) (i) $\{9, 6, 3\}$, (ii) $\{1, 2, 3\}$, (iii) $\{(1, 9), (2, 6), (3, 3)\}$; (c) $\{(3, 3)\}$.

- 2.32. $\{1, 6, 11\}, \{2, 7, 12\}, \{3, 8, 13\}, \{4, 9, 14\}, \{5, 10, 15\}$

- 2.27. (a) None; (b) (2) and (3); (c) (1) and (4); (d) all except (3).

- 2.33. (b) $\{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)\}$.

- 2.28. All are true except: (e) $R = \{(1, 2)\}, S = \{(2, 3)\}$; (f)

Functions and Algorithms

3.1 INTRODUCTION

One of the most important concepts in mathematics is that of a function. The terms “map,” “mapping,” “transformation,” and many others mean the same thing; the choice of which word to use in a given situation is usually determined by tradition and the mathematical background of the person using the term.

Related to the notion of a function is that of an algorithm. The notation for presenting an algorithm and a discussion of its complexity is also covered in this chapter.

3.2 FUNCTIONS

Suppose that to each element of a set A we assign a unique element of a set B ; the collection of such assignments is called a *function* from A into B . The set A is called the *domain* of the function, and the set B is called the *target set* or *codomain*.

Functions are ordinarily denoted by symbols. For example, let f denote a function from A into B . Then we

write

$$f: A \rightarrow B$$

which is read: “ f is a function from A into B ,” or “ f takes (or maps) A into B .” If $a \in A$, then $f(a)$ (read: “ f of a ”) denotes the unique element of B which f assigns to a ; it is called the *image* of a under f , or the *value* of f at a . The set of all image values is called the *range* or *image* of f . The image of $f: A \rightarrow B$ is denoted by $\text{Ran}(f)$, $\text{Im}(f)$ or $f(A)$.

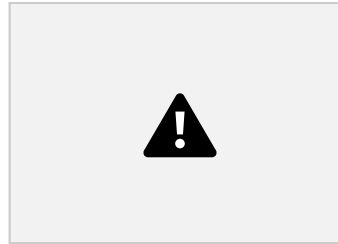
Frequently, a function can be expressed by means of a mathematical formula. For example, consider the function which sends each real number into its square. We may describe this function by writing

$$f(x) = x^2 \text{ or } x \rightarrow x^2 \text{ or } y = x^2$$

In the first notation, x is called a *variable* and the letter f denotes the function. In the second notation, the barred arrow $\rightarrow$ is read “goes into.” In the last notation, x is called the *independent variable* and y is called the *dependent variable* since the value of y will depend on the value of x .

Remark: Whenever a function is given by a formula in terms of a variable x , we assume, unless it is otherwise stated, that the domain of the function is $\mathbf{R}$ (or the largest subset of $\mathbf{R}$ for which the formula has meaning) and the codomain is $\mathbf{R}$.

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Fig. 3-1

EXAMPLE 3.1

(a) Consider the function $f(x) = x^3$, i.e., f assigns to each real number its cube. Then the image of 2 is 8, and so we may write $f(2) = 8$.

(b) Figure 3-1 defines a function f from $A = \{a, b, c, d\}$ into $B = \{r, s, t, u\}$ in the obvious way. Here f

$$(a) = s, f(b) = u, f(c) = r, f(d) = s$$

The image of f is the set of image values, $\{r, s, u\}$. Note that t does not belong to the image of f because t is not the image of any element under f .

(c) Let A be any set. The function from A into A which assigns to each element in A the element itself is called the *identity function* on A and it is usually denoted by 1_A , or simply 1. In other words, for every $a \in A$,

$$1_A(a) = a.$$

(d) Suppose S is a subset of A , that is, suppose $S \subseteq A$. The *inclusion map* or *embedding* of S into A , denoted by $i: S \rightarrow A$ is the function such that, for every $x \in S$,

$$i(x) = x$$

The *restriction* of any function $f: A \rightarrow B$, denoted by $f|_S$ is the function from S into B such that, for any $x \in S$,

$$f|_S(x) = f(x)$$

Functions as Relations

There is another point of view from which functions may be considered. First of all, every function $f: A \rightarrow B$ gives rise to a relation from A to B called the *graph* of f and defined by

$$\text{Graph of } f = \{(a, b) \mid a \in A, b = f(a)\}$$

Two functions $f: A \rightarrow B$ and $g: A \rightarrow B$ are defined to be *equal*, written $f = g$, if $f(a) = g(a)$ for every $a \in A$;

that is, if they have the same graph. Accordingly, we do not distinguish between a function and its graph. Now, such a graph relation has the property that each a in A belongs to a unique ordered pair (a, b) in the relation. On the other hand, any relation f from A to B that has this property gives rise to a function $f: A \rightarrow B$, where $f(a) = b$ for each (a, b) in f . Consequently, one may equivalently define a function as follows:

Definition: A function $f: A \rightarrow B$ is a relation from A to B (i.e., a subset of $A \times B$) such that each $a \in A$ belongs to a unique ordered pair (a, b) in f .

Although we do not distinguish between a function and its graph, we will still use the terminology “graph of f ” when referring to f as a set of ordered pairs. Moreover, since the graph of f is a relation, we can draw its picture as was done for relations in general, and this pictorial representation is itself sometimes called the graph of f . Also, the defining condition of a function, that each $a \in A$ belongs to a unique pair (a, b) in f , is equivalent to the geometrical condition of each vertical line intersecting the graph in exactly one point.

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EXAMPLE 3.2

(a) Let $f: A \rightarrow B$ be the function defined in Example 3.1 (b). Then the graph of f is as follows:

$$\{(a, s), (b, u), (c, r), (d, s)\}$$

(b) Consider the following three relations on the set $A = \{1, 2, 3\}$:

$$f = \{(1, 3), (2, 3), (3, 1)\}, g = \{(1, 2), (3, 1)\}, h = \{(1, 3), (2, 1), (1, 2), (3, 1)\}$$

f is a function from A into A since each member of A appears as the first coordinate in exactly one ordered pair in f ; here $f(1) = 3, f(2) = 3$, and $f(3) = 1$. g is not a function from A into A since $2 \in A$ is not the first coordinate of any pair in g and so g does not assign any image to 2. Also h is not a function from A into A since $1 \in A$ appears as the first coordinate of two distinct ordered pairs in h , $(1, 3)$ and $(1, 2)$. If h is to be a function it cannot assign both 3 and 2 to the element $1 \in A$.

(c) By a *real polynomial function*, we mean a function $f: \mathbf{R} \rightarrow \mathbf{R}$ of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where the a_i are real numbers. Since $\mathbf{R}$ is an infinite set, it would be impossible to plot each point of the graph. However, the graph of such a function can be approximated by first plotting some of its points and then drawing a smooth curve through these points. The points are usually obtained from a table where various values are assigned to x and the corresponding values of $f(x)$ are computed. Figure 3-2 illustrates this technique using the function $f(x) = x^2 - 2x - 3$.

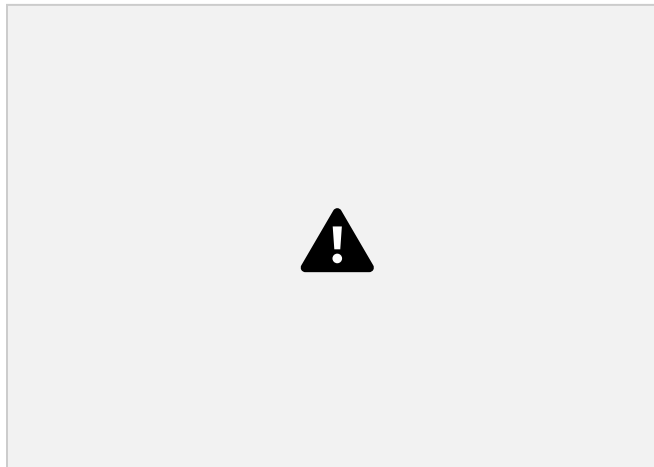


Fig. 3-2

Composition Function

Consider functions $f: A \rightarrow B$ and $g: B \rightarrow C$; that is, where the codomain of f is the domain of g . Then we may define a new function from A to C , called the *composition* of f and g and written $g \circ f$, as follows:

$$(g \circ f)(a) \equiv g(f(a))$$

That is, we find the image of a under f and then find the image of $f(a)$ under g . This definition is not really new. If we view f and g as relations, then this function is the same as the composition of f and g as relations (see Section 2.6) except that here we use the functional notation $g \circ f$ for the composition of f and g instead of the notation $f \circ g$ which was used for relations.

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Consider any function $f: A \rightarrow B$. Then

$$f \circ 1_A = f \text{ and } 1_B \circ f = f$$

where 1_A and 1_B are the identity functions on A and B , respectively.

3.3 ONE-TO-ONE, ONTO, AND INVERTIBLE FUNCTIONS

A function $f: A \rightarrow B$ is said to be *one-to-one* (written 1-1) if different elements in the domain A have

distinct images. Another way of saying the same thing is that f is *one-to-one* if $f(a) = f(a')$ implies $a = a'$. A function $f: A \rightarrow B$ is said to be an *onto* function if each element of B is the image of some element of A . In other words, $f: A \rightarrow B$ is onto if the image of f is the entire codomain, i.e., if $f(A) = B$. In such a case we say that f is a function from A onto B or that f maps A onto B .

A function $f: A \rightarrow B$ is *invertible* if its inverse relation f^{-1} is a function from B to A . In general, the inverse relation f^{-1} may not be a function. The following theorem gives simple criteria which tells us when it is.

Theorem 3.1: A function $f: A \rightarrow B$ is invertible if and only if f is both one-to-one and onto.

If $f: A \rightarrow B$ is one-to-one and onto, then f is called a *one-to-one correspondence* between A and B . This terminology comes from the fact that each element of A will then correspond to a unique element of B and vice versa.

Some texts use the terms *injective* for a one-to-one function, *surjective* for an onto function, and *bijective* for a one-to-one correspondence.

EXAMPLE 3.3 Consider the functions $f_1: A \rightarrow B$, $f_2: B \rightarrow C$, $f_3: C \rightarrow D$ and $f_4: D \rightarrow E$ defined by the diagram of Fig. 3-3. Now f_1 is one-to-one since no element of B is the image of more than one element of A . Similarly, f_2 is one-to-one. However, neither f_3 nor f_4 is one-to-one since $f_3(r) = f_3(u)$ and $f_4(v) = f_4(w)$

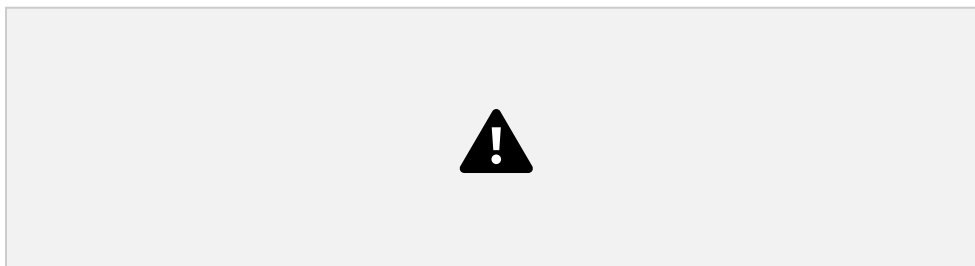


Fig. 3-3

As far as being onto is concerned, f_2 and f_3 are both onto functions since every element of C is the image under f_2 of some element of B and every element of D is the image under f_3 of some element of C , $f_2(B) = C$ and $f_3(C) = D$. On the other hand, f_1 is not onto since $3 \in B$ is not the image under f_4 of any element of A , and f_4 is not onto since $x \in E$ is not the image under f_4 of any element of D .

Thus f_1 is one-to-one but not onto, f_3 is onto but not one-to-one and f_4 is neither one-to-one nor onto. However, f_2 is both one-to-one and onto, i.e., is a one-to-one correspondence between A and B . Hence f_2 is invertible and f^{-1}

f_2 is a function from C to B .

Geometrical Characterization of One-to-One and Onto Functions

Consider now functions of the form $f: \mathbf{R} \rightarrow \mathbf{R}$. Since the graphs of such functions may be plotted in the Cartesian plane $\mathbf{R}^2$ and since functions may be identified with their graphs, we might wonder

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whether the concepts of being one-to-one and onto have some geometrical meaning. The answer is yes. Specifically:

- (1) $f: \mathbf{R} \rightarrow \mathbf{R}$ is one-to-one if each horizontal line intersects the graph of f in at most one point. (2) $f: \mathbf{R} \rightarrow \mathbf{R}$ is an onto function if each horizontal line intersects the graph of f at one or more points.

Accordingly, if f is both one-to-one and onto, i.e. invertible, then each horizontal line will intersect the graph of f at exactly one point.

EXAMPLE 3.4 Consider the following four functions from $\mathbf{R}$ into $\mathbf{R}$:

$$f_1(x) = x^2, f_2(x) = 2^x, f_3(x) = x^3 - 2x^2 - 5x + 6, f_4(x) = x^3$$

The graphs of these functions appear in Fig. 3-4. Observe that there are horizontal lines which intersect the graph of f_1 twice and there are horizontal lines which do not intersect the graph of f_1 at all; hence f_1 is neither one-to-one nor onto. Similarly, f_2 is one-to-one but not onto, f_3 is onto but not one-to-one and f_4 is both one-to-one

and onto. The inverse of f_4 is the cube root function, i.e., $f^{-1}(x) = \sqrt[3]{x}$.

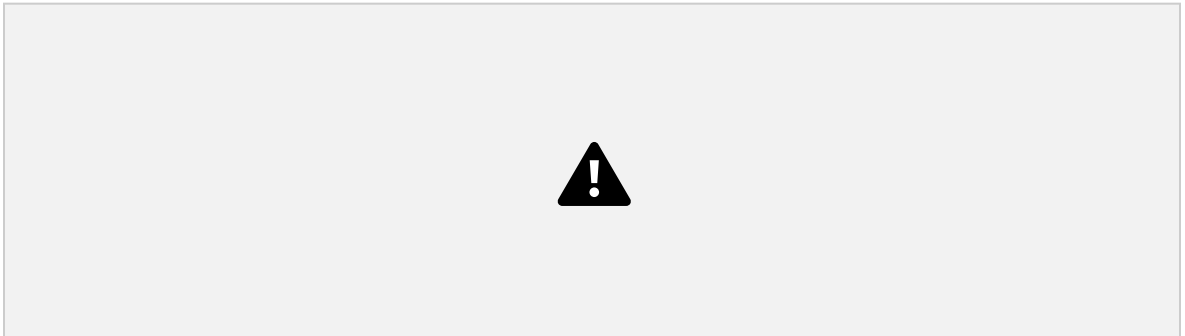


Fig. 3-4

Permutations

An invertible (bijective) function $\sigma: X \rightarrow X$ is called a *permutation* on X . The composition and inverses of permutations on X and the identity function on X are also permutations on X .

Suppose $X = \{1, 2, \dots, n\}$. Then a permutation σ on X is frequently denoted by

$$\sigma = \begin{matrix} & i_1 & i_2 & i_3 & \cdots & i_n \\ 1 & 2 & 3 & \cdots & n \end{matrix}$$

where $j_i = \sigma(i)$. The set of all such permutations is denoted by S_n , and there are $n! = n(n-1) \cdots 3 \cdot 2 \cdot 1$ of them. For example, $\sigma =$

$$123456 \quad 462513$$

$$123456 \quad 643125$$

and $\tau =$

are permutations in S_6 , and there are $6! = 720$ of them. Sometimes, we only write the second line of the permutation, that is, we denote the above permutations by writing $\sigma = 462513$ and $\tau = 643125$.

3.4 MATHEMATICAL FUNCTIONS, EXPONENTIAL AND LOGARITHMIC FUNCTIONS

This section presents various mathematical functions which appear often in the analysis of algorithms, and in computer science in general, together with their notation. We also discuss the exponential and logarithmic functions, and their relationship.

Floor and Ceiling Functions

Let x be any real number. Then x lies between two integers called the floor and the ceiling of x . Specifically,

$\lfloor x \rfloor$, called the *floor* of x , denotes the greatest integer that does not exceed x .

$\lceil x \rceil$, called the *ceiling* of x , denotes the least integer that is not less than x .

If x is itself an integer, then $\lfloor x \rfloor = x$; otherwise $\lfloor x \rfloor + 1 = \lceil x \rceil$. For example,

$$\begin{aligned} \lfloor 3.14 \rfloor &= 3, \lceil 3.14 \rceil = 4, & \lfloor -4 \rfloor &= -4, \lceil \sqrt{5} \rceil = 3, \lfloor -8.5 \rfloor = -9, \lceil 7 \rceil = 7, \\ & & \lfloor -4 \rfloor &= -4 \\ \lfloor \sqrt{5} \rfloor &= 2, \lceil -8.5 \rceil = -8, \lfloor 7 \rfloor = 7, \lceil -4 \rceil = -3 \end{aligned}$$

Integer and Absolute Value Functions

Let x be any real number. The *integer value* of x , written $\text{INT}(x)$, converts x into an integer by deleting (truncating) the fractional part of the number. Thus

$$\text{INT}(3.14) = 3, \text{INT}(\sqrt{5}) = 2, \text{INT}(-8.5) = -8, \text{INT}(7) = 7$$

Observe that $\text{INT}(x) = \lfloor x \rfloor$ or $\text{INT}(x) = \lceil x \rceil$ according to whether x is positive or negative.

The *absolute value* of the real number x , written $\text{ABS}(x)$ or $|x|$, is defined as the greater of x or $-x$. Hence $\text{ABS}(0) = 0$, and, for $x \neq 0$, $\text{ABS}(x) = x$ or $\text{ABS}(x) = -x$, depending on whether x is positive or negative. Thus

$$|-15| = 15, |7| = 7, |-3.33| = 3.33, |4.44| = 4.44, |-0.075| = 0.075$$

We note that $|x| = |-x|$ and, for $x \neq 0$, $|x|$ is positive.

Remainder Function and Modular Arithmetic

Let k be any integer and let M be a positive integer. Then

$$k \pmod{M}$$

(read: k modulo M) will denote the integer remainder when k is divided by M . More exactly, $k \pmod{M}$ is the unique integer r such that

$$k = Mq + r \text{ where } 0 \leq r < M$$

When k is positive, simply divide k by M to obtain the remainder r . Thus

$$25 \pmod{7} = 4, 25 \pmod{5} = 0, 35 \pmod{11} = 2, 3 \pmod{8} = 3$$

If k is negative, divide $|k|$ by M to obtain a remainder r ; then $k \pmod{M} = M - r$ when $r \neq 0$. Thus

$$-26 \pmod{7} = 7 - 5 = 2, -371 \pmod{8} = 8 - 3 = 5, -39 \pmod{3} = 0$$

The term “mod” is also used for the mathematical congruence relation, which is denoted and defined as follows:

$$a \equiv b \pmod{M} \text{ if and only if } M \text{ divides } b - a$$

M is called the *modulus*, and $a \equiv b \pmod{M}$ is read “ a is congruent to b modulo M ”. The following aspects of the congruence relation are frequently useful:

$$0 \equiv M \pmod{M} \text{ and } a \pm M \equiv a \pmod{M}$$

Arithmetic modulo M refers to the arithmetic operations of addition, multiplication, and subtraction where the arithmetic value is replaced by its equivalent value in the set

$$\{0, 1, 2, \dots, M-1\} \text{ or in the set } \{1, 2, 3, \dots, M\}$$

For example, in arithmetic modulo 12, sometimes called “clock” arithmetic,

$$6 + 9 \equiv 3, 7 \times 5 \equiv 11, 1 - 5 \equiv 8, 2 + 10 \equiv 0 \equiv 12$$

(The use of 0 or M depends on the application.)

Exponential Functions

Recall the following definitions for integer exponents (where m is a positive integer):

$$a^m = a \cdot a \cdots a (m \text{ times}), a^0 = 1, a^{-m} = \frac{1}{a^m}$$

Exponents are extended to include all rational numbers by defining, for any rational number m/n ,

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

For example,

$$2^4 = 16, 2^{-4} = \frac{1}{2^4} = \frac{1}{16}, 125^{2/3} = 5^2 = 25$$

In fact, exponents are extended to include all real numbers by defining, for any real number x ,

$$a^x = \lim$$

$$_{r \rightarrow x} a^r, \text{ where } r \text{ is a rational number}$$

Accordingly, the exponential function $f(x) = a^x$ is defined for all real numbers.

Logarithmic Functions

Logarithms are related to exponents as follows. Let b be a positive number. The logarithm of any positive number x to the base b , written

$$\log_b x$$

represents the exponent to which b must be raised to obtain x . That is,

$$y = \log_b x \text{ and } b^y = x$$

are equivalent statements. Accordingly,

$$\begin{aligned} \log_2 8 &= 3 \text{ since } 2^3 = 8; \log_{10} 100 = 2 \text{ since } 10^2 = 100 \\ \log_2 64 &= 6 \text{ since } 2^6 = 64; \log_{10} 0.001 = -3 \text{ since } 10^{-3} = 0.001 \end{aligned}$$

Furthermore, for any base b , we have $b^0 = 1$ and $b^1 = b$; hence

$$\log_b 1 = 0 \text{ and } \log_b b = 1$$

The logarithm of a negative number and the logarithm of 0 are not defined.

Frequently, logarithms are expressed using approximate values. For example, using tables or calculators, one obtains

$$\log_{10} 300 = 2.4771 \text{ and } \log_e 40 = 3.6889$$

as approximate answers. (Here $e = 2.718281\dots$)

Three classes of logarithms are of special importance: logarithms to base 10, called *common logarithms*; logarithms to base e , called *natural logarithms*; and logarithms to base 2, called *binary logarithms*. Some texts write

$\ln x$ for $\log_e x$ and $\lg x$ or $\log x$ for $\log_2 x$

The term $\log x$, by itself, usually means $\log_{10} x$; but it is also used for $\log_e x$ in advanced mathematical texts and for $\log_2 x$ in computer science texts.

Frequently, we will require only the floor or the ceiling of a binary logarithm. This can be obtained by looking at the powers of 2. For example,

$$\log_2 100 = 6 \text{ since } 2^6 = 64 \text{ and } 2^7 = 128$$

$$\log_2 1000 = 9 \text{ since } 2^8 = 512 \text{ and } 2^9 = 1024$$

and so on.

Relationship between the Exponential and Logarithmic Functions

The basic relationship between the exponential and the logarithmic functions

$$f(x) = b^x \text{ and } g(x) = \log_b x$$

is that they are inverses of each other; hence the graphs of these functions are related geometrically. This relationship is illustrated in Fig. 3-5 where the graphs of the exponential function $f(x) = 2^x$, the logarithmic function $g(x) = \log_2 x$, and the linear function $h(x) = x$ appear on the same coordinate axis. Since $f(x) = 2^x$ and $g(x) = \log_2 x$ are inverse functions, they are symmetric with respect to the linear function $h(x) = x$ or, in other words, the line $y = x$.

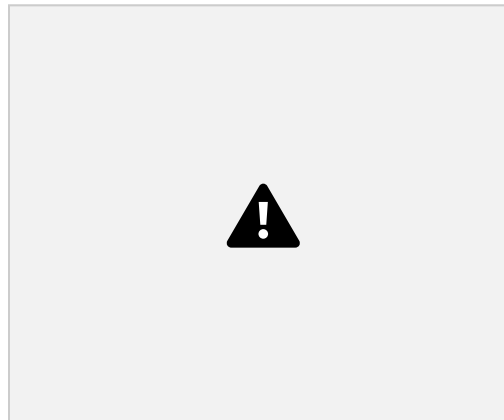


Fig. 3-5

Figure 3-5 also indicates another important property of the exponential and logarithmic functions. Specifically, for any positive c , we have

$$g(c) < h(c) < f(c), \text{ that is, } g(c) < c < f(c)$$

In fact, as c increases in value, the vertical distances $h(c) - g(c)$ and $f(c) - g(c)$ increase in value. Moreover, the logarithmic function $g(x)$ grows very slowly compared with the linear function $h(x)$, and the exponential function $f(x)$ grows very quickly compared with $h(x)$.

3.5 SEQUENCES, INDEXED CLASSES OF SETS

Sequences and indexed classes of sets are special types of functions with their own notation. We discuss these objects in this section. We also discuss the summation notation here.

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Sequences

A *sequence* is a function from the set $\mathbf{N} = \{1, 2, 3, \dots\}$ of positive integers into a set A . The notation a_n is used to denote the image of the integer n . Thus a sequence is usually denoted by

$$a_1, a_2, a_3, \dots \text{ or } \{a_n; n \in \mathbf{N}\} \text{ or simply } \{a_n\}$$

Sometimes the domain of a sequence is the set $\{0, 1, 2, \dots\}$ of nonnegative integers rather than $\mathbf{N}$. In such a case we say n begins with 0 rather than 1.

A *finite sequence* over a set A is a function from $\{1, 2, \dots, m\}$ into A , and it is usually denoted by

$$a_1, a_2, \dots, a_m$$

Such a finite sequence is sometimes called a *list* or an *m-tuple*.

EXAMPLE 3.5

(a) The following are two familiar sequences:

(i) $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ which may be defined by $a_n = \frac{1}{n}$;

(ii) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ which may be defined by $b_n = 2^{-n}$

Note that the first sequence begins with $n = 1$ and the second sequence begins with $n = 0$.

(b) The important sequence $1, -1, 1, -1, \dots$ may be formally defined by

$$a_n = (-1)^{n+1} \text{ or, equivalently, by } b_n = (-1)^n$$

where the first sequence begins with $n = 1$ and the second sequence begins with $n = 0$.

(c) **Strings** Suppose a set A is finite and A is viewed as a character set or an alphabet. Then a finite sequence over A is called a *string* or *word*, and it is usually written in the form $a_1 a_2 \dots a_m$, that is, without parentheses. The number m of characters in the string is called its *length*. One also views the set with zero characters as a string; it is called the *empty string* or *null string*. Strings over an alphabet A and certain operations on these strings will be discussed in detail in Chapter 13.

Summation Symbol, Sums

Here we introduce the summation symbol (the Greek letter sigma). Consider a sequence $a_1, a_2, a_3, \dots$. Then we define the following:

$$\sum_{j=1}^n a_j = a_1 + a_2 + \dots + a_n \quad \sum_{j=m}^n a_j = a_m + a_{m+1} + \dots + a_n$$

The letter j in the above expressions is called a *dummy index* or *dummy variable*. Other letters frequently used as dummy variables are i, k, s , and t .

$$\sum_{i=1}^n a_i b_i = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

EXAMPLE 3.6

$$\sum_{j=2}^5 j^2 = 2^2 + 3^2 + 4^2 + 5^2 = 54$$

$$\sum_{j=1}^n j = 1 + 2 + \dots + n$$

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The last sum appears very often. It has the value $n(n+1)/2$. That is:

$$1 + 2 + 3 + \dots + n = n(n+1)/2$$

$$2, \text{ for example, } 1 + 2 + \dots + 50 = 50(51)/2$$

$$= 1275$$

Indexed Classes of Sets

Let I be any nonempty set, and let S be a collection of sets. An *indexing function* from I to S is a function $f: I \rightarrow S$. For any $i \in I$, we denote the image $f(i)$ by A_i . Thus the indexing function f is usually denoted by

$$\{A_i \mid i \in I\} \text{ or } \{A_i\}_{i \in I} \text{ or simply } \{A_i\}$$

The set I is called the *indexing set*, and the elements of I are called *indices*. If f is one-to-one and onto, we say that S is indexed by I .

The concepts of union and intersection are defined for indexed classes of sets as follows:

$$\bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\} \text{ and } \bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}$$

In the case that I is a finite set, this is just the same as our previous definition of union and intersection. If I is $\mathbf{N}$, we may denote the union and intersection, respectively, as follows:

$$A_1 \cup A_2 \cup A_3 \cup \dots \text{ and } A_1 \cap A_2 \cap A_3 \cap \dots$$

EXAMPLE 3.7 Let I be the set $\mathbf{Z}$ of integers. To each $n \in \mathbf{Z}$, we assign the following infinite interval in $\mathbf{R}$:

$$A_n = \{x \mid x \leq n\} = (-\infty, n]$$

For any real number a , there exists integers n_1 and n_2 such that $n_1 < a < n_2$; so $a \in A_{n_2}$ but $a \notin A_{n_1}$. Hence a

$$\in \bigcup_n A_n \text{ but } a \notin \bigcap_n A_n$$

Accordingly,

$$\bigcup_n A_n = \mathbf{R} \text{ but } \bigcap_n A_n = \emptyset$$

3.6 RECURSIVELY DEFINED FUNCTIONS

A function is said to be *recursively defined* if the function definition refers to itself. In order for the definition not to be circular, the function definition must have the following two properties:

- (1) There must be certain arguments, called *base values*, for which the function does not refer to itself.
- (2) Each time the function does refer to itself, the argument of the function must be closer to a base value.

A recursive function with these two properties is said to be *well-defined*.

The following examples should help clarify these ideas.

Factorial Function

The product of the positive integers from 1 to n , inclusive, is called “ n factorial” and is usually denoted by $n!$. That is,

$$n! = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1$$

It is also convenient to define $0! = 1$, so that the function is defined for all nonnegative integers. Thus:

$$0! = 1, 1! = 1, 2! = 2 \cdot 1 = 2, 3! = 3 \cdot 2 \cdot 1 = 6, 4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120, 6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

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And so on. Observe that

$$5! = 5 \cdot 4! = 5 \cdot 24 = 120 \text{ and } 6! = 6 \cdot 5! = 6 \cdot 120 = 720$$

This is true for every positive integer n ; that is,

$$n! = n \cdot (n-1)!$$

Accordingly, the factorial function may also be defined as follows:

Definition 3.1 (Factorial Function):

- (a) If $n = 0$, then $n! = 1$.
- (b) If $n > 0$, then $n! = n \cdot (n-1)!$

Observe that the above definition of $n!$ is recursive, since it refers to itself when it uses $(n-1)!$. However:

- (1) The value of $n!$ is explicitly given when $n = 0$ (thus 0 is a base value).
- (2) The value of $n!$ for arbitrary n is defined in terms of a smaller value of n which is closer to the base value 0.

Accordingly, the definition is not circular, or, in other words, the function is well-defined.

EXAMPLE 3.8 Figure 3-6 shows the nine steps to calculate $4!$ using the recursive definition. Specifically:

Step 1. This defines $4!$ in terms of $3!$, so we must postpone evaluating $4!$ until we evaluate 3. This postponement is indicated by indenting the next step.

Step 2. Here $3!$ is defined in terms of $2!$, so we must postpone evaluating $3!$ until we evaluate

$2!$. **Step 3.** This defines $2!$ in terms of $1!$.

Step 4. This defines $1!$ in terms of $0!$.

Step 5. This step can explicitly evaluate $0!$, since 0 is the base value of the recursive definition.

Steps 6 to 9. We backtrack, using $0!$ to find $1!$, using $1!$ to find $2!$, using $2!$ to find $3!$, and finally using $3!$ to find $4!$. This backtracking is indicated by the “reverse” indention.

Observe that we backtrack in the reverse order of the original postponed evaluations. **Fig. 3-6**

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Level Numbers

Let P be a procedure or recursive formula which is used to evaluate $f(X)$ where f is a recursive function and X is the input. We associate a *level number* with each execution of P as follows. The original execution of P is assigned level 1; and each time P is executed because of a recursive call, its level is one more than the level of the execution that made the recursive call. The *depth* of recursion in evaluating $f(X)$ refers to the maximum level number of P during its execution.

Consider, for example, the evaluation of $4!$ Example 3.8, which uses the recursive formula $n! = n(n - 1)!$. Step 1 belongs to level 1 since it is the first execution of the formula. Thus:

Step 2 belongs to level 2; Step 3 to level 3,...; Step 5 to level 5.

On the other hand, Step 6 belongs to level 4 since it is the result of a return from level 5. In other words, Step 6 and Step 4 belong to the same level of execution. Similarly,

Step 7 belongs to level 3; Step 8 to level 2; and Step 9 to level 1.

Accordingly, in evaluating $4!$, the depth of the recursion is 5.

Fibonacci Sequence

The celebrated Fibonacci sequence (usually denoted by $F_0, F_1, F_2, \dots$) is as follows:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

That is, $F_0 = 0$ and $F_1 = 1$ and each succeeding term is the sum of the two preceding terms. For example, the next two terms of the sequence are

$$34 + 55 = 89 \text{ and } 55 + 89 = 144$$

A formal definition of this function follows:

Definition 3.2 (Fibonacci Sequence):

- (a) If $n = 0$, or $n = 1$, then $F_n = n$.
- (b) If $n > 1$, then $F_n = F_{n-2} + F_{n-1}$.

This is another example of a recursive definition, since the definition refers to itself when it uses F_{n-2} and F_{n-1} . However:

- (1) The base values are 0 and 1.
- (2) The value of F_n is defined in terms of smaller values of n which are closer to the base values.

Accordingly, this function is well-defined.

Ackermann Function

The Ackermann function is a function with two arguments, each of which can be assigned any nonnegative integer, that is, 0, 1, 2, This function is defined as:

Definition 3.3 (Ackermann function):

- (a) If $m = 0$, then $A(m, n) = n + 1$.
- (b) If $m = 0$ but $n = 0$, then $A(m, n) = A(m - 1, 1)$.
- (c) If $m = 0$ and $n = 0$, then $A(m, n) = A(m - 1, A(m, n - 1))$.

Once more, we have a recursive definition, since the definition refers to itself in parts (b) and (c). Observe that $A(m, n)$ is explicitly given only when $m = 0$. The base criteria are the pairs

$$(0, 0), (0, 1), (0, 2), (0, 3), \dots, (0, n), \dots$$

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Although it is not obvious from the definition, the value of any $A(m, n)$ may eventually be expressed in terms of the value of the function on one or more of the base pairs.

The value of $A(1, 3)$ is calculated in Problem 3.21. Even this simple case requires 15 steps. Generally speaking, the Ackermann function is too complex to evaluate on any but a trivial example. Its importance comes from its use in mathematical logic. The function is stated here mainly to give another example of a classical recursive function and to show that the recursion part of a definition may be complicated.

3.7 CARDINALITY

Two sets A and B are said to be *equipotent*, or to have the *same number of elements* or the *same cardinality*, written $A \sim B$, if there exists a one-to-one correspondence $f: A \rightarrow B$. A set A is *finite* if A is empty or if A has the same cardinality as the set $\{1, 2, \dots, n\}$ for some positive integer n . A set is *infinite* if it is not finite. Familiar examples of infinite sets are the natural numbers $\mathbf{N}$, the integers $\mathbf{Z}$, the rational numbers $\mathbf{Q}$, and the real numbers $\mathbf{R}$.

We now introduce the idea of “cardinal numbers”. We will consider cardinal numbers simply as symbols assigned to sets in such a way that two sets are assigned the same symbol if and only if they have the same cardinality. The cardinal number of a set A is commonly denoted by $|A|$, $n(A)$, or $\text{card}(A)$. We will use $|A|$.

The obvious symbols are used for the cardinality of finite sets. That is, 0 is assigned to the empty set $\emptyset$, and n is assigned to the set $\{1, 2, \dots, n\}$. Thus $|A| = n$ if and only if A has n elements. For example,

$$|\{x, y, z\}| = 3 \text{ and } |\{1, 3, 5, 7, 9\}| = 5$$

The cardinal number of the infinite set $\mathbf{N}$ of positive integers is $\aleph_0$ (“aleph-naught”). This symbol was introduced by Cantor. Thus $|A| = \aleph_0$ if and only if A has the same cardinality as $\mathbf{N}$.

EXAMPLE 3.9 Let $E = \{2, 4, 6, \dots\}$, the set of even positive integers. The function $f: \mathbf{N} \rightarrow E$ defined by $f(n) = 2n$ is a one-to-one correspondence between the positive integers $\mathbf{N}$ and E . Thus E has the same cardinality as $\mathbf{N}$ and so we may write

$$|E| = \aleph_0$$

A set with cardinality $\aleph_0$ is said to be *denumerable* or *countably infinite*. A set which is finite or denumerable is said to be *countable*. One can show that the set $\mathbf{Q}$ of rational numbers is countable. In fact, we have the following theorem (proved in Problem 3.13) which we will use subsequently.

Theorem 3.2: A countable union of countable sets is countable.

That is, if $A_1, A_2, \dots$ are each countable sets, then the following union is countable:

$$A_1 \cup A_2 \cup A_3 \cup \dots$$

An important example of an infinite set which is uncountable, i.e., not countable, is given by the following theorem which is proved in Problem 3.14.

Theorem 3.3: The set $\mathbf{I}$ of all real numbers between 0 and 1 is uncountable.

Inequalities and Cardinal Numbers

One also wants to compare the size of two sets. This is done by means of an inequality relation which is defined for cardinal numbers as follows. For any sets A and B , we define $|A| \leq |B|$ if there exists a function $f: A \rightarrow B$ which is one-to-one. We also write

$$|A| < |B| \text{ if } |A| \leq |B| \text{ but } |A| \neq |B|$$

For example, $|\mathbf{N}| < |\mathbf{I}|$, where $\mathbf{I} = \{x: 0 \leq x \leq 1\}$, since the function $f: \mathbf{N} \rightarrow \mathbf{I}$ defined by $f(n) = 1/n$ is one-to-one, but $|\mathbf{N}| \neq |\mathbf{I}|$ by Theorem 3.3.

Cantor's Theorem, which follows and which we prove in Problem 3.25, tells us that the cardinal numbers are unbounded.

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Theorem 3.4 (Cantor): For any set A , we have $|A| < |\text{Power}(A)|$ (where $\text{Power}(A)$ is the power set of A , i.e., the collection of all subsets of A).

The next theorem tells us that the inequality relation for cardinal numbers is antisymmetric.

Theorem 3.5: (Schroeder-Bernstein): Suppose A and B are sets such that

$$|A| \leq |B| \text{ and } |B| \leq |A|$$

Then $|A| = |B|$.

We prove an equivalent formulation of this theorem in Problem 3.26.

3.8 ALGORITHMS AND FUNCTIONS

An algorithm M is a finite step-by-step list of well-defined instructions for solving a particular problem, say, to find the output $f(X)$ for a given function f with input X . (Here X may be a list or set of values.) Frequently, there may be more than one way to obtain $f(X)$, as illustrated by the following examples. The particular choice of the algorithm M to obtain $f(X)$ may depend on the "efficiency" or "complexity" of the algorithm; this question of the complexity of an algorithm M is formally discussed in the next section.

EXAMPLE 3.10 (Polynomial Evaluation) Suppose, for a given polynomial $f(x)$ and value $x = a$, we want to find $f(a)$, say,

$$f(x) = 2x^3 - 7x^2 + 4x - 15 \text{ and } a = 5$$

This can be done in the following two ways.

(a) (**Direct Method**): Here we substitute $a = 5$ directly in the polynomial to obtain $f(5) =$

$$2(125) - 7(25) + 4(5) - 7 = 250 - 175 + 20 - 15 = 80$$

Observe that there are $3 + 2 + 1 = 6$ multiplications and 3 additions. In general, evaluating a polynomial of degree n directly would require approximately

$$n + (n - 1) + \cdots + 1 = n(n + 1)$$

2 multiplications and n additions.

- (b) (**Horner's Method or Synthetic Division**): Here we rewrite the polynomial by successively factoring out x (on the right) as follows:

$$f(x) = (2x^2 - 7x + 4)x - 15 = ((2x - 7)x + 4)x - 15$$

Then

$$f(5) = ((3)5 + 4)5 - 15 = (19)5 - 15 = 95 - 15 = 80$$

For those familiar with synthetic division, the above arithmetic is equivalent to the following synthetic division:

$$\begin{array}{r} 5 \quad 2 \quad -7 \quad +4 \quad -15 \\ \hline \end{array}$$

$$10 + 15 + 95$$

$$2 + 3 + 19 + 80$$

Observe that here there are 3 multiplications and 3 additions. In general, evaluating a polynomial of degree n by Horner's method would require approximately

n multiplications and n additions

Clearly Horner's method (b) is more efficient than the direct method (a).

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EXAMPLE 3.11 (Greatest Common Divisor) Let a and b be positive integers with, say, $b < a$; and suppose we want to find $d = \text{GCD}(a, b)$, the greatest common divisor of a and b . This can be done in the following two ways.

- (a) (**Direct Method**): Here we find all the divisors of a , say by testing all the numbers from 2 to $a/2$, and all the divisors of b . Then we pick the largest common divisor. For example, suppose $a = 258$ and $b = 60$. The divisors of a and b follow:

$a = 258$; divisors: 1, 2, 3, 6, 86, 129, 258

$b = 60$; divisors: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60 Accordingly, $d = \text{GCD}(258, 60)$

$= 6$.

- (b) (**Euclidean Algorithm**): Here we divide a by b to obtain a remainder r_1 . (Note $r_1 < b$.) Then we divide b by the remainder r_1 to obtain a second remainder r_2 . (Note $r_2 < r_1$.) Next we divide r_1 by r_2 to obtain a third remainder r_3 . (Note $r_3 < r_2$.) We continue dividing r_k by r_{k+1} to obtain a remainder r_{k+2} . Since

$$a > b > r_1 > r_2 > r_3 \dots (*)$$

eventually we obtain a remainder $r_m = 0$. Then $r_{m-1} = \text{GCD}(a, b)$. For example, suppose $a = 258$ and $b = 60$. Then:

(1) Dividing $a = 258$ by $b = 60$ yields the remainder $r_1 = 18$.

(2) Dividing $b = 60$ by $r_1 = 18$ yields the remainder $r_2 = 6$.

(3) Dividing $r_1 = 18$ by $r_2 = 6$ yields the remainder $r_3 = 0$.

Thus $r_2 = 6 = \text{GCD}(258, 60)$.

The Euclidean algorithm is a very efficient way to find the greatest common divisor of two positive integers a and b . The fact that the algorithm ends follows from (*). The fact that the algorithm yields $d = \text{GCD}(a, b)$ is not obvious; it is discussed in Section 11.6.

3.9 COMPLEXITY OF ALGORITHMS

The analysis of algorithms is a major task in computer science. In order to compare algorithms, we must have some criteria to measure the efficiency of our algorithms. This section discusses this important topic. Suppose M is an algorithm, and suppose n is the size of the input data. The time and space used by the algorithm are the two main measures for the efficiency of M . The time is measured by counting the number of “key operations;” for example:

- (a) In sorting and searching, one counts the number of comparisons.
- (b) In arithmetic, one counts multiplications and neglects additions.

Key operations are so defined when the time for the other operations is much less than or at most proportional to the time for the key operations. The space is measured by counting the maximum of memory needed by the algorithm.

The *complexity* of an algorithm M is the function $f(n)$ which gives the running time and/or storage space requirement of the algorithm in terms of the size n of the input data. Frequently, the storage space required by an algorithm is simply a multiple of the data size. Accordingly, unless otherwise stated or implied, the term “complexity” shall refer to the running time of the algorithm.

The complexity function $f(n)$, which we assume gives the running time of an algorithm, usually depends not only on the size n of the input data but also on the particular data. For example, suppose we want to search through an English short story TEXT for the first occurrence of a given 3-letter word W . Clearly, if W is the 3-letter word “the,” then W likely occurs near the beginning of TEXT, so $f(n)$ will be small. On the other hand, if W is the 3-letter word “zoo,” then W may not appear in TEXT at all, so $f(n)$ will be large.

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The above discussion leads us to the question of finding the complexity function $f(n)$ for certain cases. The two cases one usually investigates in complexity theory are as follows:

- (1) *Worst case*: The maximum value of $f(n)$ for any possible input.
- (2) *Average case*: The expected value of $f(n)$.

The analysis of the average case assumes a certain probabilistic distribution for the input data; one possible assumption might be that the possible permutations of a data set are equally likely. The average case also uses the following concept in probability theory. Suppose the numbers $n_1, n_2, \dots, n_k$ occur with respective probabilities $p_1, p_2, \dots, p_k$. Then the *expectation* or *average value* E is given by

$$E = n_1p_1 + n_2p_2 + \dots + n_kp_k$$

These ideas are illustrated below.

Linear Search

Suppose a linear array DATA contains n elements, and suppose a specific ITEM of information is given. We want either to find the location LOC of ITEM in the array DATA, or to send some message, such as LOC = 0, to indicate that ITEM does not appear in DATA. The linear search algorithm solves this problem by comparing ITEM, one by one, with each element in DATA. That is, we compare ITEM with DATA[1], then DATA[2], and so on, until we find LOC such that ITEM = DATA[LOC].

The complexity of the search algorithm is given by the number C of comparisons between ITEM and DATA[K]. We seek $C(n)$ for the worst case and the average case.

- (1) **Worst Case**: Clearly the worst case occurs when ITEM is the last element in the array DATA or is not there at all. In either situation, we have

$$C(n) = n$$

Accordingly, $C(n) = n$ is the worst-case complexity of the linear search algorithm.

- (2) **Average Case**: Here we assume that ITEM does appear in DATA, and that it is equally likely to occur at any position in the array. Accordingly, the number of comparisons can be any of the numbers 1, 2, 3, ..., n , and each number occurs with probability $p = 1/n$. Then

$$\begin{aligned}
C(n) &= 1 \cdot \frac{1}{n} + 2 \cdot \frac{1}{n} + \cdots + n \cdot \frac{1}{n} \\
&= (1 + 2 + \cdots + n) \cdot \frac{1}{n} \\
&= \frac{1}{2} \cdot n = n + 1 \\
&= n(n + 1) \\
&2
\end{aligned}$$

This agrees with our intuitive feeling that the average number of comparisons needed to find the location of ITEM is approximately equal to half the number of elements in the DATA list.

Remark: The complexity of the average case of an algorithm is usually much more complicated to analyze than that of the worst case. Moreover, the probabilistic distribution that one assumes for the average case may not actually apply to real situations. Accordingly, unless otherwise stated or implied, the complexity of an algorithm shall mean the function which gives the running time of the worst case in terms of the input size. This is not too strong an assumption, since the complexity of the average case for many algorithms is proportional to the worst case.

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Rate of Growth; Big O Notation

Suppose M is an algorithm, and suppose n is the size of the input data. Clearly the complexity $f(n)$ of M increases as n increases. It is usually the rate of increase of $f(n)$ that we want to examine. This is usually done by comparing $f(n)$ with some standard function, such as

$$\log n, n, n \log n, n^2, n^3, 2^n$$

The rates of growth for these standard functions are indicated in Fig. 3-7, which gives their approximate values for certain values of n . Observe that the functions are listed in the order of their rates of growth: the logarithmic function $\log_2 n$ grows most slowly, the exponential function 2^n grows most rapidly, and the polynomial functions n^c grow according to the exponent c .

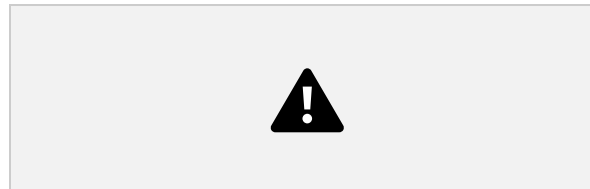


Fig. 3-7 Rate of growth of standard functions

The way we compare our complexity function $f(n)$ with one of the standard functions is to use the functional “big O ” notation which we formally define below.

Definition 3.4: Let $f(x)$ and $g(x)$ be arbitrary functions defined on $\mathbf{R}$ or a subset of $\mathbf{R}$. We say “ $f(x)$ is of order $g(x)$,” written

$$f(x) = O(g(x))$$

if there exists a real number k and a positive constant C such that, for all $x > k$, we have $|f$

$$(x)| \leq C|g(x)|$$

In other words, $f(x) = O(g(x))$ if a constant multiple of $|g(x)|$ exceeds $|f(x)|$ for all x greater than some real number k .

We also write:

$$f(x) = h(x) + O(g(x)) \text{ when } f(x) - h(x) = O(g(x))$$

(The above is called the “big O ” notation since $f(x) = o(g(x))$ has an entirely different meaning.)

Consider now a polynomial $P(x)$ of degree m . We show in Problem 3.24 that $P(x) = O(x^m)$. Thus, for example,

$$7x^2 - 9x + 4 = O(x^2) \text{ and } 8x^3 - 576x^2 + 832x - 248 = O(x^3)$$

Complexity of Well-known Algorithms

Assuming $f(n)$ and $g(n)$ are functions defined on the positive integers, then

$$f(n) = O(g(n))$$

means that $f(n)$ is bounded by a constant multiple of $g(n)$ for almost all n .

To indicate the convenience of this notation, we give the complexity of certain well-known searching and sorting algorithms in computer science:

- (a) Linear search: $O(n)$ (c) Bubble sort: $O(n^2)$
 (b) Binary search: $O(\log n)$ (d) Merge-sort: $O(n \log n)$

60 FUNCTIONS AND ALGORITHMS [CHAP. 3] Solved Problems

FUNCTIONS

3.1. Let $X = \{1, 2, 3, 4\}$. Determine whether each relation on X is a function from X into X .

- (a) $f = \{(2, 3), (1, 4), (2, 1), (3, 2), (4, 4)\}$
 (b) $g = \{(3, 1), (4, 2), (1, 1)\}$
 (c) $h = \{(2, 1), (3, 4), (1, 4), (2, 1), (4, 4)\}$

Recall that a subset f of $X \times X$ is a function $f: X \rightarrow X$ if and only if each $a \in X$ appears as the first coordinate in exactly one ordered pair in f .

- (a) No. Two different ordered pairs $(2, 3)$ and $(2, 1)$ in f have the same number 2 as their first coordinate.
 (b) No. The element $2 \in X$ does not appear as the first coordinate in any ordered pair in g .
 (c) Yes. Although $2 \in X$ appears as the first coordinate in two ordered pairs in h , these two ordered pairs are equal.

3.2. Sketch the graph of: (a) $f(x) = x^2 + x - 6$; (b) $g(x) = x^3 - 3x^2 - x + 3$.

Set up a table of values for x and then find the corresponding values of the function. Since the functions are polynomials, plot the points in a coordinate diagram and then draw a smooth continuous curve through the points. See Fig. 3-8.

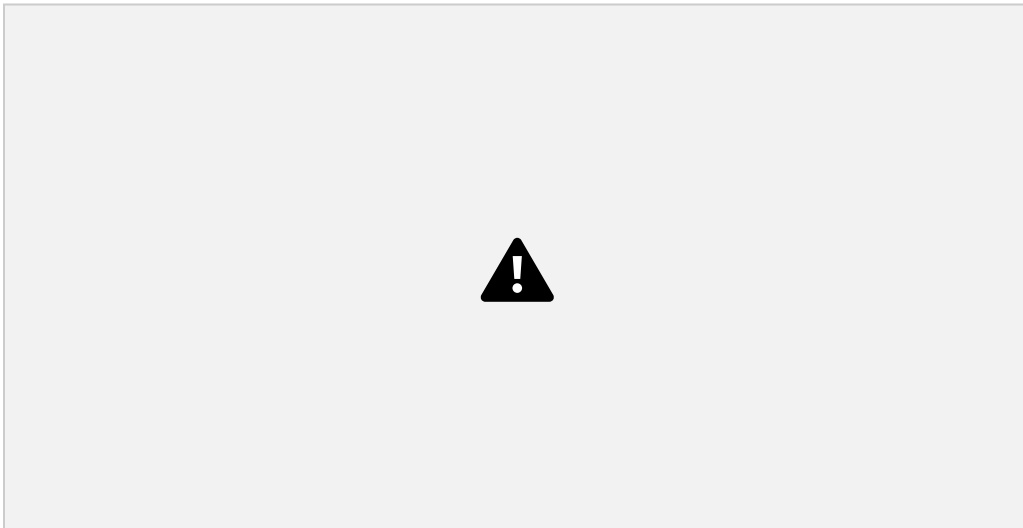


Fig. 3-8

3.3. Let $A = \{a, b, c\}$, $B = \{x, y, z\}$, $C = \{r, s, t\}$. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be defined by: $f =$

$$\{(a, y), (b, x), (c, y)\} \text{ and } g = \{(x, s), (y, t), (z, r)\}.$$

Find: (a) composition function $g \circ f: A \rightarrow C$; (b) $\text{Im}(f)$, $\text{Im}(g)$, $\text{Im}(g \circ f)$.

- (a) Use the definition of the composition function to compute:

$$(g \circ f)(a) = g(f(a)) = g(y) = t$$

$$(g \circ f)(b) = g(f(b)) = g(x) = s$$

$$(g \circ f)(c) = g(f(c)) = g(y) = t$$

That is $g \circ f = \{(a, t), (b, s), (c, t)\}$.

(b) Find the image points (or second coordinates):

$$\text{Im}(f) = \{x, y\}, \text{Im}(g) = \{r, s, t\}, \text{Im}(g \circ f) = \{s, t\}$$

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3.4. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and $g: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = 2x + 1$ and $g(x) = x^2 - 2$. Find the formula for the composition function $g \circ f$.

Compute $g \circ f$ as follows: $(g \circ f)(x) = g(f(x)) = g(2x + 1) = (2x + 1)^2 - 2 = 4x^2 + 4x - 1$. Observe that the same answer can be found by writing

$$y = f(x) = 2x + 1 \text{ and } z = g(y) = y^2 - 2$$

and then eliminating y from both equations:

$$z = y^2 - 2 = (2x + 1)^2 - 2 = 4x^2 + 4x - 1$$

ONE-TO-ONE, ONTO, AND INVERTIBLE FUNCTIONS

3.5. Let the functions $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$ be defined by Fig. 3-9. Determine if each function is: (a) onto, (b) one-to-one, (c) invertible.

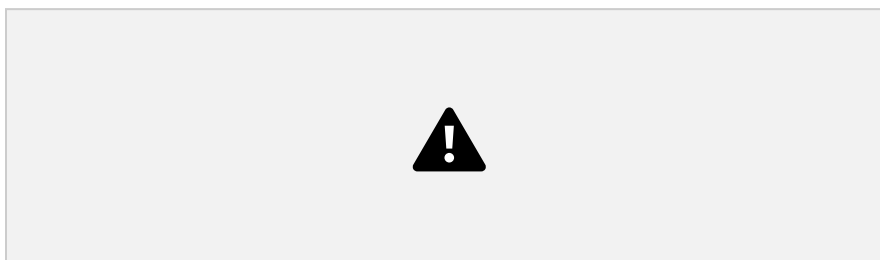


Fig. 3-9

- (a) The function $f: A \rightarrow B$ is not onto since $3 \in B$ is not the image of any element in A .
 The function $g: B \rightarrow C$ is not onto since $z \in C$ is not the image of any element in B .
 The function $h: C \rightarrow D$ is onto since each element in D is the image of some element of C .
- (b) The function $f: A \rightarrow B$ is not one-to-one since a and c have the same image 2.
 The function $g: B \rightarrow C$ is one-to-one since 1, 2 and 3 have distinct images.
 The function $h: C \rightarrow D$ is not one-to-one since x and z have the same image 4.
- (c) No function is one-to-one and onto; hence no function is invertible.

permutations $\sigma =$

123456 246531 in S_6 .

3.6. Consider 123456 364512 and $\tau =$

Find: (a) composition $\tau \circ \sigma$; (b) σ^{-1} .

(a) Note that σ sends 1 into 3 and τ sends 3 into 6. So the composition $\tau \circ \sigma$ sends 1 into 6. I.e. $(\tau \circ \sigma)(1) = 6$. Moreover, $\tau \circ \sigma$ sends 2 into 6 into 1 that is, $(\tau \circ \sigma)(2) = 1$. Similarly,

$$(\tau \circ \sigma)(3) = 5, (\tau \circ \sigma)(4) = 3, (\tau \circ \sigma)(5) = 2, (\tau \circ \sigma)(6) = 4$$

Thus

$$123456 \rightarrow 615324$$

$$\tau \circ \sigma =$$

(b) Look for 1 in the second row of σ . Note σ sends 5 into 1. Hence $\sigma^{-1}(1) = 5$. Look for 2 in the second row of σ . Note σ sends 6 into 2. Hence $\sigma^{-1}(2) = 6$. Similarly, $\sigma^{-1}(3) = 1$, $\sigma^{-1}(4) = 3$, $\sigma^{-1}(5) = 4$, $\sigma^{-1}(6) = 2$. Thus

$$\sigma^{-1} = 564321$$

3.7. Consider functions $f: A \rightarrow B$ and $g: B \rightarrow C$. Prove the following:

- (a) If f and g are one-to-one, then the composition function $g \circ f$ is one-to-one.
 (b) If f and g are onto functions, then $g \circ f$ is an onto function.
 (a) Suppose $(g \circ f)(x) = (g \circ f)(y)$; then $g(f(x)) = g(f(y))$. Hence $f(x) = f(y)$ because g is one-to-one. Further more, $x = y$ since f is one-to-one. Accordingly $g \circ f$ is one-to-one.
 (b) Let c be any arbitrary element of C . Since g is onto, there exists a $b \in B$ such that $g(b) = c$. Since f is onto, there exists an $a \in A$ such that $f(a) = b$. But then

$$(g \circ f)(a) = g(f(a)) = g(b) = c$$

Hence each $c \in C$ is the image of some element $a \in A$. Accordingly, $g \circ f$ is an onto function.

3.8. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = 2x - 3$. Now f is one-to-one and onto; hence f has an inverse function f^{-1} . Find a formula for f^{-1} .

Let y be the image of x under the function f :

$$y = f(x) = 2x - 3$$

Consequently, x will be the image of y under the inverse function f^{-1} . Solve for x in terms of y in the above equation:

$$x = (y + 3)/2$$

Then $f^{-1}(y) = (y + 3)/2$. Replace y by x to obtain

$$f^{-1}(x) = \frac{x + 3}{2}$$

which is the formula for f^{-1} using the usual independent variable x .

3.9. Prove the following generalization of DeMorgan's law: For any class of sets $\{A_i\}$ we have

$$(\cup_i A_i)^c = \cap_i A_i^c$$

We have:

$$x \in (\cup_i A_i)^c \text{ iff } x \notin \cup_i A_i \text{ iff } \forall_i \in I, x \notin A_i \text{ iff } \forall_i \in I, x \in A_i^c \text{ iff } x \in \cap_i A_i^c$$

Therefore, $(\cup_i A_i)^c = \cap_i A_i^c$. (Here we have used the logical notations iff for "if and only" if and $\forall$ for "for all.")

CARDINALITY

3.10. Find the cardinal number of each set:

- (a) $A = \{a, b, c, \dots, y, z\}$, (c) $C = \{10, 20, 30, 40, \dots\}$,
 (b) $B = \{x \mid x \in \mathbf{N}, x^2 = 5\}$, (d) $D = \{6, 7, 8, 9, \dots\}$.
 (a) $|A| = 26$ since there are 26 letters in the English alphabet.
 (b) $|B| = 0$ since there is no positive integer whose square is 5, that is, B is empty.
 (c) $|C| = \aleph_0$ because $f: \mathbf{N} \rightarrow C$, defined by $f(n) = 10n$, is a one-to-one correspondence between $\mathbf{N}$ and C . (d)
 $|D| = \aleph_0$ because $g: \mathbf{N} \rightarrow D$, defined by $g(n) = n + 5$ is a one-to-one correspondence between $\mathbf{N}$ and D .

3.11. Show that the set $\mathbf{Z}$ of integers has cardinality $\aleph_0$.

The following diagram shows a one-to-one correspondence between $\mathbf{N}$ and $\mathbf{Z}$:

$$\begin{array}{ccccccccccc} \mathbf{N} = & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & \dots & \\ & & & & & & & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \dots \\ \mathbf{Z} = & 0 & 1 & -1 & 2 & -2 & 3 & -3 & 4 & \dots \end{array}$$

That is, the following function $f: \mathbf{N} \rightarrow \mathbf{Z}$ is one-to-one and onto:

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ (1 - n)/2 & \text{if } n \text{ is odd} \end{cases}$$

Accordingly, $|\mathbf{Z}| = |\mathbf{N}| = \aleph_0$.

3.12. Let $A_1, A_2, \dots$ be a countable number of finite sets. Prove that the union $S = \bigcup_i A_i$ is countable.

Essentially, we list the elements of A_1 , then we list the elements of A_2 which do not belong to A_1 , then we list the elements of A_3 which do not belong to A_1 or A_2 , i.e., which have not already been listed, and so on. Since the A_i are finite, we can always list the elements of each set. This process is done formally as follows.

First we define sets $B_1, B_2, \dots$ where B_j contains the elements of A_j which do not belong to preceding sets, i.e., we define

$$B_1 = A_1 \text{ and } B_k = A_k \setminus (A_1 \cup A_2 \cup \dots \cup A_{k-1})$$

Then the B_j are disjoint and $S = \bigcup_j B_j$. Let $b_{i1}, b_{i2}, \dots, b_{im}$ be the elements of B_i . Then $S = \{b_{ij}\}$. Let $f: S \rightarrow \mathbb{N}$ be defined as follows:

$$f(b_{ij}) = m_1 + m_2 + \dots + m_{j-1} + j$$

If S is finite, then S is countable. If S is infinite then f is a one-to-one correspondence between S and $\mathbb{N}$. Thus S is countable.

3.13. Prove Theorem 3.2: A countable union of countable sets is countable.

Suppose $A_1, A_2, A_3, \dots$ are a countable number of countable sets. In particular, suppose $a_{i1}, a_{i2}, a_{i3}, \dots$ are the elements of A_i . Define sets $B_2, B_3, B_4, \dots$ as follows:

$$B_k = \{a_{ij} \mid i + j = k\}$$

For example, $B_6 = \{a_{15}, a_{24}, a_{33}, a_{42}, a_{51}\}$. Observe that each B_k is finite and

$$S = \bigcup_i A_i = \bigcup_k B_k$$

By the preceding problem $\bigcup_k B_k$ is countable. Hence $S = \bigcup_i A_i$ is countable and the theorem is proved.

3.14. Prove Theorem 3.3: The set $\mathbf{I}$ of all real numbers between 0 and 1 inclusive is uncountable. The set $\mathbf{I}$ is clearly infinite, since it contains $1, \frac{1}{2}, \frac{1}{3}, \dots$. Suppose $\mathbf{I}$ is denumerable. Then there exists a one-to-one correspondence $f: \mathbb{N} \rightarrow \mathbf{I}$. Let $f(1) = a_1, f(2) = a_2, \dots$; that is, $\mathbf{I} = \{a_1, a_2, a_3, \dots\}$. We list the elements $a_1, a_2, \dots$ in a column and express each in its decimal expansion:

$$\begin{array}{l} a_1 = 0.x_{11}x_{12}x_{13}x_{14} \dots \\ a_2 = 0.x_{21}x_{22}x_{23}x_{24} \dots \\ a_3 = 0.x_{31}x_{32}x_{33}x_{34} \dots \\ a_4 = 0.x_{41}x_{42}x_{43}x_{44} \dots \\ \dots \end{array}$$

where $x_{ij} \in \{0, 1, 2, \dots, 9\}$. (For those numbers which can be expressed in two different decimal expansions, e.g., $0.2000000 \dots = 0.1999999 \dots$, we choose the expansion which ends with nines.)

Let $b = 0.y_1y_2y_3y_4 \dots$ be the real number obtained as follows:

$$y_i = \begin{array}{ll} 1 & \text{if } x_{ii} = 2 \\ 2 & \text{if } x_{ii} = 1 \\ 0 & \text{if } x_{ii} \neq 1, 2 \end{array}$$

Now $b \in \mathbf{I}$. But $b = a_1$ because $y_1 = x_{11}$

$$b = a_2 \text{ because } y_2 = x_{22}$$

$$b = a_3 \text{ because } y_3 = x_{33}$$

.....

Therefore b does not belong to $\mathbf{I} = \{a_1, a_2, \dots\}$. This contradicts the fact that $b \in \mathbf{I}$. Hence the assumption that $\mathbf{I}$ is denumerable must be false, so $\mathbf{I}$ is uncountable.

SPECIAL MATHEMATICAL FUNCTIONS

3.15. Find: (a) 7.5 , -7.5 , -18 ; (b) 7.5 , -7.5 , -18 .

(a) By definition, x denotes the greatest integer that does not exceed x , hence $7.5 = 7$, $-7.5 = -8$, $-18 = -18$.

3.16. Find: (a) $25 \pmod{7}$; (b) $25 \pmod{5}$; (c) $-35 \pmod{11}$; (d) $-3 \pmod{8}$.

When k is positive, simply divide k by the modulus M to obtain the remainder r . Then $r = k \pmod{M}$. If k is negative, divide $|k|$ by M to obtain the remainder r . Then $k \pmod{M} = M - r$ (when $r \neq 0$). Thus:

$$(a) 25 \pmod{7} = 4 \quad (c) -35 \pmod{11} = 11 - 2 = 9$$

$$(b) 25 \pmod{5} = 0 \quad (d) -3 \pmod{8} = 8 - 3 = 5$$

3.17. Evaluate modulo $M = 15$: (a) $9 + 13$; (b) $7 + 11$; (c) $4 - 9$; (d) $2 - 10$.

Use $a \pm M = a \pmod{M}$:

$$(a) 9 + 13 = 22 = 22 - 15 = 7 \quad (c) 4 - 9 = -5 = -5 + 15 = 10$$

$$(b) 7 + 11 = 18 = 18 - 15 = 3 \quad (d) 2 - 10 = -8 = -8 + 15 = 7$$

3.18. Simplify: (a) $n!$

$$(n-1)!; (b) (n+2)!$$

$$n! \cdot$$

$$(n-1)! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$$

$$(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1 = n \text{ or, simply, } n!$$

$$(a) n! \qquad n! = (n+2)(n+1)n!$$

$$(b) (n+2)! \qquad (n-1)! = n(n-1)! \quad (n-1)! = n$$

$$n! = (n+2)(n-1) = n^2 + 3n + 2$$

3.19. Evaluate: (a) $\log_2 8$; (b) $\log_2 64$; (c) $\log_{10} 100$; (d) $\log_{10} 0.001$.

$$(a) \log_2 8 = 3 \text{ since } 2^3 = 8 \quad (c) \log_{10} 100 = 2 \text{ since } 10^2 = 100 \quad (b) \log_2 64 = 6 \text{ since } 2^6 = 64 \quad (d) \log_{10} 0.001 = -3 \text{ since } 10^{-3} = 0.001$$

RECURSIVE FUNCTIONS

3.20. Let a and b be positive integers, and suppose Q is defined recursively as follows:

$$Q(a, b) = \begin{cases} 0 & \text{if } a < b \\ Q(a-b, b) + 1 & \text{if } b \leq a \end{cases}$$

(a) Find: (i) $Q(2, 5)$; (ii) $Q(12, 5)$.

(b) What does this function Q do? Find $Q(5861, 7)$.

(a) (i) $Q(2, 5) = 0$ since $2 < 5$.

$$\begin{aligned} \text{(ii) } Q(12, 5) &= Q(7, 5) + 1 \\ &= [Q(2, 5) + 1] + 1 = Q(2, 5) + 2 \\ &= 0 + 2 = 2 \end{aligned}$$

(b) Each time b is subtracted from a , the value of Q is increased by 1. Hence $Q(a, b)$ finds the quotient when a is divided by b . Thus $Q(5861, 7) = 837$.

3.21. Use the definition of the Ackermann function to find $A(1, 3)$.

Figure 3-10 shows the 15 steps that are used to evaluate $A(1, 3)$.

The forward indentation indicates that we are postponing an evaluation and are recalling the definition, and the backward indentation indicates that we are backtracking. Observe that (a) of the definition is used in Steps 5, 8, 11 and 14; (b) in Step 4; and (c) in Steps 1, 2, and 3. In the other steps we are backtracking with substitutions.

Fig. 3-10

MISCELLANEOUS PROBLEMS

3.22. Find the domain D of each of the following real-valued functions of a real variable:

$$(c) f(x) = \sqrt{25 - x^2}$$

$$(a) f(x) = \frac{1}{x-2}$$

$$(b) f(x) = x^2 - 3x - 4 \quad (d) x^2 \text{ where } 0 \leq x \leq 2$$

When a real-valued function of a real variable is given by a formula $f(x)$, then the domain D consists of the largest subset of $\mathbf{R}$ for which $f(x)$ has meaning and is real, unless otherwise specified.

(a) f is not defined for $x - 2 = 0$, i.e., for $x = 2$; hence $D = \mathbf{R} \setminus \{2\}$.

(b) f is defined for every real number; hence $D = \mathbf{R}$.

(c) f is not defined when $25 - x^2$ is negative; hence $D = [-5, 5] = \{x \mid -5 \leq x \leq 5\}$.

(d) Here, the domain of f is explicitly given as $D = \{x \mid 0 \leq x \leq 2\}$.

3.23. For any $n \in \mathbf{N}$, let $D_n = (0, 1/n)$, the open interval from 0 to $1/n$. Find:

$$(a) D_3 \cup D_4; (b) D_3 \cap D_{20}; (c) D_s \cup D_t; (d) D_s \cap D_t.$$

(a) Since $(0, 1/3)$ is a superset of $(0, 1/7)$, $D_3 \cup D_4 = D_3$.

(b) Since $(0, 1/20)$ is a subset of $(0, 1/3)$, $D_3 \cap D_{20} = D_{20}$.

(c) Let $m = \min(s, t)$, that is, the smaller of the two numbers s and t ; then D_m is equal to D_s or D_t contains the other as a subset. Hence $D_s \cap D_t = D_m$.

(d) Let $M = \max(s, t)$, that is, the larger of the two numbers s and t ; then $D_s \cap D_t = D_M$.

3.24. Suppose $P(n) = a_0 + a_1n + a_2n^2 + \cdots + a_mn^m$ has degree m . Prove $P(n) = O(n^m)$.

$$\text{Let } b_0 = |a_0|, b_1 = |a_1|, \dots, b_m = |a_m|.$$

$$\text{Then for } n \geq 1, p(n) \leq b_0 + b_1n + b_2n^2 + \cdots + b_mn^m = \underline{b_0} \leq (b_0 + b_1 + \cdots + b_m)n^m = Mn^m$$

$$+ \cdots + b_mn^m = \underline{b_0}$$

$$\text{where } M = |a_0| + |a_1| + \cdots + |a_m|. \text{ Hence } P(n) = O(n^m).$$

$$\frac{m}{n} + \frac{b_1}{1}$$

$$\frac{m-1}{n} + \cdots + b_m$$

$$\text{For example, } 5x^3 + 3x = O(x^3) \text{ and } x^5 - 4000000x^2 = O(x^5).$$

3.25. Prove Theorem 3.4 (Cantor): $|A| < |\text{Power}(A)|$ (where $\text{Power}(A)$ is the power set of A). The function $g: A \rightarrow \text{Power}(A)$ defined by $g(a) = \{a\}$ is clearly one-to-one; hence $|A| \leq |\text{Power}(A)|$.

If we show that $|A| = |\text{Power}(A)|$, then the theorem will follow. Suppose the contrary, that is, suppose $|A| < |\text{Power}(A)|$ and that $f: A \rightarrow \text{Power}(A)$ is a function which is both one-to-one and onto. Let $a \in A$ be called a “bad” element if $a \notin f(a)$, and let B be the set of bad elements. In other words,

$$B = \{x : x \in A, x \notin f(x)\}$$

Now B is a subset of A . Since $f : A \rightarrow \text{Power}(A)$ is onto, there exists $b \in A$ such that $f(b) = B$. Is b a “bad” element or a “good” element? If $b \in B$ then, by definition of B , $b \notin f(b) = B$, which is impossible. Likewise, if $b \notin B$ then $b \in f(b) = B$, which is also impossible. Thus the original assumption that $|A| = |\text{Power}(A)|$ has led to a contradiction. Hence the assumption is false, and so the theorem is true.